

Kidney & Upper-Tract Tumours

Rapid Revision

Campbell-Walsh-Wein · Chapters 136-139, 141-144

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total under 5 h even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we...?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: underlined bold = number to memorise · **MCQ** = exam trivia only · → *Ch 141* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
136	Benign Renal Tumours	21 min
137	Malignant Renal Tumours	56 min
138	Urothelial Tumours of the Upper Tract & Ureter	28 min
139	Surgical Management of UTUC	33 min
141	Contemporary Open Surgery of the Kidney	33 min
142	Laparoscopic & Robotic Surgery of the Kidney	32 min
143	Focal Therapy for Renal Tumours	21 min
144	Treatment of Advanced RCC	36 min
★	Putting it together: the big treatment choices	10 min
Total (18946 words, at 70 words/min)		4h 30min

136 Benign Renal Tumours

Reading time ≈ 21 min (slow pace)

1 • THE BENIGN SMALL RENAL MASS

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Ultrasonography (US), computed tomography (CT) and magnetic resonance imaging (MRI) cannot reliably separate benign from renal cell carcinoma (RCC). Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are diagnosable on imaging. Small size, female sex and older age are unreliable pointers.
- Management follows RCC risk: no follow-up (simple cyst) → selective arterial embolisation (SAE) (large AML) → excision (solid or complex cystic mass). Renal mass biopsy and the SRM algorithm → Ch 137.

2 • RENAL CYSTS AND THE BOSNIAK CLASSIFICATION

- Commonest renal mass (↑ with age, male sex, renal dysfunction); usually asymptomatic; grow 1–2 mm/yr.
- Simple cyst**: any size, homogeneous, **≤20 Hounsfield units (HU)** → no further imaging. Heterogeneity, >20 HU, septa, wall thickening, nodules or calcification → Bosniak staging.
- Version 2019** added MRI criteria, mainly because too many resected Bosniak III lesions proved benign.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (≤2 mm) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) ≤3 thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (3 mm) enhancing wall/septum, or ≥4 thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to 5 yr
III	Enhancing thick (≥4 mm) or irregular (≤3 mm obtuse-margined protrusion) wall/septum	~50%	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	~90%	

KEY POINT Memory ladder: wall ≤2 → 3 → ≥4 mm → irregularity → nodule. Traps: any calcification is compatible with II; a Bosniak I wall may enhance; large (≥3 cm) hyperdense cysts are no longer IIF. Bosniak I/II growth is expected and not a sign of malignancy.

KEY POINT Why surveil small III/IV? Bosniak III has a low positive predictive value, and cystic RCC is indolent: much of its volume is fluid, so tumour burden is low and size predicts outcome poorly (proposed stage **pT1c**). The number needed to treat to prevent one metastasis is therefore very high.

- European Association of Urology (EAU): manage Bosniak IV as a localised solid mass. Biopsy only if a significant solid component and the result would change management. Thermal ablation for cystic masses: effectiveness uncertain.

Diffuse cystic disease

	Autosomal dominant polycystic kidney disease (ADPKD)	Acquired cystic kidney disease (ACKD)
Cause	PKD1 (chr 16, polycystin-1); PKD2 (chr 4, polycystin-2) — ciliary calcium signalling	End-stage renal disease (ESRD) on dialysis; cysts from proximal tubule; 80% by 10 yr
Kidneys	Grossly enlarged; a cause of ESRD	Small/normal; a consequence of ESRD
RCC	Risk not raised, but earlier, more sarcomatoid, harder to diagnose	Risk ≥10x ; multifocal, bilateral, male; papillary commoner; distinct World Health Organization (WHO) 2022 entity

- ADPKD: hypertension (renin-aldosterone, ischaemia); extrarenal cysts (liver, pancreas, spleen, ovary, seminal vesicle, arachnoid). US diagnosis in PKD1 relatives aged 15-29: **≥3 cysts**; a normal scan excludes disease reliably only from age 40.
- Nephrological care: blood pressure control, lifestyle; **tolvaptan** (vasopressin V2 antagonist) slows decline in rapid progressors.
- Refer to urology for stones, flank pain, gross haematuria, cyst haemorrhage (self-limiting in days) or cyst infection (fever, raised inflammatory markers; confirm with fluorodeoxyglucose positron emission tomography (FDG-PET); empirical fluoroquinolone or trimethoprim-sulfamethoxazole).
- Intervention for bulk pain only: aspiration, embolisation, fenestration; nephrectomy in selected patients only (morbidity). ESRD → transplantation.

3 · ONCOCYTOMA

- **Commonest benign enhancing renal mass**; up to half of benign SRMs <4 cm.
- Syndromic in **Birt-Hogg-Dubé** (autosomal dominant, folliculin, chr 17): facial fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC). Renal oncocytosis (diffuse bilateral nodules → chronic kidney disease, CKD) may harbour chRCC/hybrid tumours.
- Imaging: central **stellate scar** (non-specific); lower corticomedullary attenuation, gradual enhancement and **segmental enhancement inversion** favour oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive
Cytogenetics	Normal, -1, -Y, 11q13 rearrangement	Extensive abnormalities

KEY POINT Why can a core not clear oncocytoma? Sampling error, and chRCC contains oncocytoma-like areas — a substantial minority of biopsy "oncocytomas" are RCC at surgery. EAU: definitive diagnosis is not possible on a needle core; adequate tissue with CK7 and cytogenetics helps.

4 · ANGIOMYOLIPOMA

Pathology and genetics

- Perivascular epithelioid cell tumour: dysmorphic vessels, smooth muscle, fat. Sporadic (women, 4th-5th decade) or tuberous sclerosis complex (TSC; AML in **55-90%**) or lymphangioleiomyomatosis (LAM).
- Biallelic loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34). Hamartin-tuberin normally inhibit mammalian target of rapamycin (mTOR); loss → unchecked growth and angiogenesis. Multiple tumours are clonally independent.
- Immunohistochemistry: **HMB-45**, **Melan-A**, smooth muscle actin positive; keratin negative.

	Sporadic	TSC-related	Epithelioid
Lesions	Solitary, smaller	Bilateral, multifocal, recurrent; haemorrhage, hypertension, CKD	Minimal fat; aggressive (necrosis, venous extension, metastasis)

	Sporadic	TSC-related	Epithelioid
Surveillance	US/CT at 6 and 12 months, then yearly	MRI every 1-3 yr	—
Treatment	If triggers met (below)	Everolimus if <u>>3 cm</u>	As RCC

Diagnosis

- Macroscopic fat (**-15 to -20 HU** on non-contrast CT) is diagnostic. US unreliable (some RCCs hyperechoic). MRI: signal loss on fat suppression, India-ink artefact. Fat-poor AML mimics RCC → MRI/biopsy.
- Giant AML arises from parenchyma; retroperitoneal liposarcoma envelops and compresses the kidney.

Management

ALGORITHM

Spontaneous retroperitoneal haemorrhage (Wunderlich†) → resuscitate → **SAE** → nephrectomy only if SAE unavailable or unsuitable

Symptomatic (pain, haematuria, barriers to care) → SAE → nephron-sparing surgery (NSS) or RN → ablation in selected cases

Asymptomatic sporadic: >6 cm, aneurysm ≥5 mm, growth >0.25 cm/yr, childbearing age/pregnancy → treat; otherwise surveil

TSC: >3 cm or growing → everolimus first line; SAE or kidney-sparing resection second

KEY POINT Why abandon 4 cm? Bleeding risk rises steeply with size but the 4 cm cut-off was poorly specific, so most size-treated patients were overtreated. AML <4 cm → surveillance; observed AML ≥4 cm → thin-slice CT angiography to detect aneurysm. "Barriers to care" is an indication because surveillance presumes access to imaging and emergency care. Pregnancy raises bleeding risk and stakes → consider prophylactic treatment before conception.

- SAE:** first line for acute bleeding, symptoms, prophylaxis and pre-surgery. Post-embolisation syndrome (fever, pain, leucocytosis, vomiting) is common and self-limiting; abscess and renal infarction occur; major complications rare.
- NSS:** preferred operation — preserves function, crucial in TSC because lesions keep recurring.
- TSC consensus: haemorrhage → embolise then corticosteroids; avoid nephrectomy. EAU (sporadic): mTOR inhibitor only if unsuitable for SAE/surgery.

5 · OTHER BENIGN EPITHELIAL AND MESENCHYMAL TUMOURS

- Papillary adenoma:** unencapsulated, ≤15 mm (≤5 mm until 2015); shares +7, +17, -Y and KMT2C with papillary RCC; no metastatic capacity; acceptable in a donor kidney.
- Metanephric adenoma:** women, 50s; may cause polycythaemia. CT: hypovascular, mild homogeneous enhancement — not diagnostic, and biopsy accuracy unproven → excise by NSS.
Wilms tumour 1 (WT1), CD57 positive; CK7, AMACR negative; lacks papillary RCC's +7/+17/-Y.
- Mixed epithelial and stromal tumour (MEST) and adult cystic nephroma:** one WHO 2022 family; middle-aged women; oestrogen/progesterone-receptor-positive stroma; multiseptated, mimic cystic RCC → manage by Bosniak guidelines, consider biopsy; rare malignant change. Paediatric cystic nephroma is distinct (**DICER1**).
- Leiomyoma:** from renal capsule; T1/T2 hypointense; smooth muscle actin, desmin, caldesmon positive. Often actually lipid-poor AML: both may be HMB-45 positive, but **Melan-A** is positive only in AML. NSS if surgery.

Tumour	Key trait
Haemangioma	Sturge-Weber, Klippel-Trénaunay; usually excised
Haemangioblastoma	Von Hippel-Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria; aspiration/sclerosis or excision
Juxtaglomerular cell tumour (reninoma)	Young women; refractory hypertension, hypokalaemia, high renin; ≤ 3 cm; renin granules pathognomonic; excision cures most
Solitary fibrous tumour	~10% aggressive: age ≥ 55 , ≥ 15 cm, mitoses, necrosis

6 • LANDMARK TRIAL

Trial	Design	Result	Take-home
EXIST-2	Phase III randomised, everolimus vs placebo, TSC/LAM AML	$\geq 50\%$ volume reduction 42% vs 0% ; no bleeding requiring embolisation	Approved; mTOR inhibition first line in TSC

7 • MCQ ONE-LINERS

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. Simple cyst: ≤ 20 HU. 2. Bosniak III ~50%, IV ~90% malignant. 3. ≤ 3 thin septa = II; ≥ 4 = IIF. 4. IIF: surveillance up to 5 years. 5. Proposed cystic RCC stage: pT1c. 6. PKD1 chr 16; PKD2 chr 4. 7. ACKD cysts arise from proximal tubule. 8. Everolimus: stomatitis, acne, proteinuria, hypercholesterolaemia. 9. Commonest benign enhancing SRM: oncocytoma. 10. Birt-Hogg-Dubé: folliculin, chr 17. 1. Oncocytoma: KIT positive, CK7 single-cell. 2. Segmental enhancement inversion favours oncocytoma. | <ol style="list-style-type: none"> 13. AML: HMB-45, Melan-A positive; keratin negative. 14. TSC2 tuberlin 16p13; TSC1 hamartin 9q34. 15. AML intralesional aneurysm ≥ 5 mm predicts bleeding. 16. TSC AML > 3 cm: everolimus. 17. Papillary adenoma ≤ 15 mm. 18. Metanephric adenoma: WT1, CD57 positive; polycythaemia. 19. DICER1: paediatric cystic nephroma. 20. Leiomyoma: commonest origin renal capsule. 21. Renal haemangioblastoma: VHL. 22. Reninoma: hypertension plus hypokalaemia in young woman. |
|---|--|

137 Malignant Renal Tumours

Reading time ≈ 56 min (slow pace)

1 · OVERVIEW, EPIDEMIOLOGY & AETIOLOGY

- Renal cell carcinoma (RCC) = 2-3% of adult cancers; the most lethal common urological cancer. Male:female **2:1**; peak 55-75 y; **4-6% familial**.
- Incidence ↑ ~3%/y, driven by incidental detection on ultrasound (USG) and computed tomography (CT); >60% now incidental, ~70% organ-confined. Because imaging finds many small indolent tumours, incidence has risen far faster than mortality — the basis for today's emphasis on avoiding overtreatment.
- Risk factors: **tobacco** (most accepted; relative risk ~1.4, dose-dependent, falls after quitting), **obesity** ("obesity paradox": higher body mass index but lower stage), **hypertension** (the disease, not its drugs); chronic kidney disease (CKD)/end-stage renal disease (ESRD); family history 2-4×.
- Cell of origin: clear cell & papillary → **proximal tubule**; chromophobe & collecting duct → distal nephron.
- Children/young adults: RCC 1-6% of paediatric renal tumours (Wilms ~90%), equal to Wilms in the 2nd decade; commonest = **Xp11/TFE3 translocation RCC**. They respond well to surgery even when node-positive → formal lymph node dissection (LND) at radical nephrectomy (RN) recommended.

2 · CLASSIFICATION & PATHOLOGY

World Health Organization (WHO) 2022 classification (key_groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; epithelioid angiomyolipoma ; metastases

KEY POINT Trap: epithelioid angiomyolipoma is malignant, whereas classic angiomyolipoma (AML), papillary adenoma and oncocytoma are benign. The papillary type 1/2 split is **abandoned** — "type 2" proved to be several diseases (MET-altered, MiT translocation, FH-deficient) with very different prognoses, which is why WHO moved to molecular definitions.

Gross features

- Round, with a **pseudocapsule** (compressed parenchyma, not a true capsule); not infiltrative except collecting duct and sarcomatoid.
- Venous involvement ~10% (more than any tumour), up to the right atrium. Bilateral 2-4% sporadic; multicentric 10-20%. Small satellite lesions, invisible on imaging, are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtypes

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70-80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Golden-yellow, hypervascular; clear cytoplasm (dissolved glycogen, lipid); carbonic anhydrase IX (CA-IX)+; sarcomatoid 5-8%	Most venous invasion; worse localised, better metastatic
Papillary (10-15%)	Trisomy 7, 17 , loss Y; MET altered 81%	Hypovascular; papillae on fibrovascular cores; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival 86-92%); common in acquired cystic disease
Chromophobe (3-5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in BHD	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	Intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; cup-like CA-IX	ESRD, VHL; now considered benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8+; diagnosis of exclusion	Poor; urothelial-like → may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	Sickle cell trait	Young Black patients; infiltrative; SMARCB1 loss	Dismal, median 8-13 mo; modest platinum response
FH-deficient	FH (1q42-43)	Macronucleolus with perinucleolar halo; FH loss, 2SC+	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells + papillae; fluorescence in situ hybridisation (FISH) confirms	40% of paediatric RCC; t(X;1) late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
Unclassified (1-5%)	—	Catch-all, shrinking	Poor

- **Indolent:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC, clear cell papillary.
- **Lethal:** medullary, FH-deficient, collecting duct, unclassified, sarcomatoid.
- **Sarcomatoid (1-5%):** a dedifferentiated area of any subtype, not a separate tumour; median survival <1 y. Sarcomatoid or rhabdoid change = automatic grade 4 and must be reported in every RCC.

KEY POINT Why does localised clear cell RCC do worse than papillary/chromophobe, yet metastatic clear cell does better? It is intrinsically more aggressive, but its VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/International Society of Urological Pathology (ISUP) grading — replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at ×100
4	Extreme pleomorphism, giant cells, rhabdoid/sarcomatoid

3 · MOLECULAR BIOLOGY & HEREDITARY SYNDROMES

Von Hippel-Lindau (VHL)-hypoxia-inducible factor (HIF)-vascular endothelial growth factor (VEGF)_pathway

ALGORITHM

Normoxia: VHL protein + elongin B/C + cullin-2 + RBX1 = E3 ubiquitin ligase → HIF-1α/2α degraded

VHL loss or hypoxia → **HIF-2α** accumulates → dimerises with ARNT → ↑ VEGF (angiogenesis), PDGF, TGF-α, GLUT1, erythropoietin, CA-IX → hypervascular clear cell RCC

Drug targets: VEGF ligand (bevacizumab) · VEGF receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib) · mTOR (temsirolimus, everolimus) · HIF-2α (**belzutifan**, US Food and Drug Administration (FDA) 2021 for VHL, ~50% response) · CTLA-4 (ipilimumab), PD-1 (nivolumab, pembrolizumab), PD-L1 (atezolizumab, avelumab).

- Biology explains therapeutic history: multidrug-resistance pumps → chemotherapy fails (response 5–7%); VHL-driven vascularity → anti-angiogenics work; immune evasion via PD-L1/PD-1 → checkpoint blockade is now the mainstay → *Ch 144*.
- Knudson two-hit: familial patients inherit one hit in every cell → early, bilateral, multifocal tumours. All familial RCC syndromes are **autosomal dominant**; all are tumour-suppressor losses except **MET (oncogene)**.
- Poor prognosis in all subtypes: **CDKN2A** loss, hypermethylation. BAP1 mutation → rapid growth, unsuitable for active surveillance (AS).

Hereditary RCC syndromes

Syndrome	Gene (chr)	Renal tumour	Other features	Renal management
VHL	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	central nervous system (CNS)/retinal haemangioblastoma, pheochromocytoma, pancreatic & renal cysts, endolymphatic sac tumour, epididymal cystadenoma	AS <3 cm; nephron-sparing ≥3 cm; belzutifan
Hereditary papillary RCC	MET (7q31) oncogene	Multiple bilateral papillary	None	AS <3 cm; nephron-sparing ≥3 cm
Hereditary leiomyomatosis & RCC (HLRCC)	FH (1q42–43)	Aggressive, often solitary ; collecting duct	Cutaneous & uterine leiomyomas; uterine leiomyosarcoma	Prompt excision at any size, wide margins ; PN only if wide margin feasible
SDH-RCC	SDHA/B/C/D, SDHAF2	Variable, onset ≤40 y	Paraganglioma/pheo, gastrointestinal stromal tumour (GIST), thyroid	Excision, PN preferred
Birt-Hogg-Dubé (BHD)	FLCN (17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax	AS <3 cm; nephron-sparing ≥3 cm
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast (50%), thyroid	Screen CT/MRI; 3-cm rule
Tuberous sclerosis	TSC1 (9q34)/TSC2 (16p13)	Multiple AMLs; clear cell 1–3%	Cardiac rhabdomyoma, angiofibromas, lymphangioleiomyomatosis, epilepsy	AML: surveil <3, everolimus 3–5, embolise/excise ≥5 cm

Syndrome	Gene (chr)	Renal tumour	Other features	Renal management
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma	Excision, nephron-sparing

- **VHL genotype-phenotype:** type 1 (deletion/truncation) = no pheochromocytoma; type 2 (missense) = pheochromocytoma — 2A low RCC, 2B high RCC, 2C pheochromocytoma only.
- VHL: RCC now the commonest cause of death (mean onset ~39 y).

4 • CLINICAL PRESENTATION

- **Classic triad** (flank pain, gross haematuria, mass) is rare and signals advanced disease ("too late triad").
- Advanced-disease clues: weight loss, bone pain, cervical nodes, **non-reducing or right-sided varicocele**, leg oedema (inferior vena cava [IVC] obstruction).
- Spontaneous perirenal haematoma without cause: >50% harbour a tumour (AML or RCC) → repeat CT after a few months.

Paraneoplastic syndromes (10-20%)

The kidney normally makes renin, erythropoietin, vitamin D and prostaglandins; RCC overproduces these plus parathyroid hormone-related peptide (PTHrP) and cytokines.

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Advanced disease marker
Hypercalcaemia	Up to 13%; PTHrP or osteolytic metastases	Only one manageable medically: hydration → furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy ± metastasectomy
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3-20%; interleukin-6	Non-metastatic liver dysfunction (↑ alkaline phosphatase almost always); exclude liver metastases; normalises after nephrectomy in 60-70%; persistence = viable tumour, poor prognosis

5 • DIAGNOSIS

Risk of malignancy

- Strongest predictors: **male sex (~3×)** and **size**. Benign: 20-30% of ≤4 cm, ~40% of ≤2 cm; at ≥7 cm few benign, half high grade. Clinical nomograms are little better than a coin toss (concordance 0.55-0.65).
- Clinical T1 mass: ~20% benign (oncocytoma, fat-poor AML) · ~60% indolent RCC · ~20% potentially aggressive.

Imaging

- **Multiphase CT** (without + with contrast) = single most important test: enhancement **>15-20 Hounsfield units (HU)** suggests RCC; fat **<-20 HU** = AML. Magnetic resonance imaging (MRI): enhancement >20%; best for IVC thrombus and adjacent-organ invasion; macrocyclic gadolinium is safe even on dialysis (nephrogenic systemic fibrosis now near-absent).

- USG simple cyst: smooth wall, round, anechoic, posterior enhancement — anything else needs cross-sectional imaging. Contrast-enhanced USG if CT/MRI contraindicated.
- fluorodeoxyglucose positron emission tomography (FDG-PET): limited (renal background uptake), not for staging. **Sestamibi single-photon emission CT (SPECT)/CT**: mitochondria-rich **oncocyoma**/hybrid tumours take it up. ¹²⁴I-girentuximab PET targets CA-IX (clear cell).
- Enhancement: clear cell & oncocytoma strong (overlap — cannot separate); papillary/chromophobe moderate. Infiltrative pattern → lymphoma, urothelial carcinoma, sarcomatoid, collecting duct, medullary, xanthogranulomatous pyelonephritis.
- Complexity scores — **R.E.N.A.L.** (Radius, Endophytic/exophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines), PADUA, C-index — guide PN vs RN, approach and complication risk.

ALGORITHM

Simple cyst → observe

Complex/solid: estimated glomerular filtration rate (eGFR) >45 → renal CT or MRI with/without contrast · eGFR 30–45 → hydrate, same · eGFR <30 → non-contrast CT or gadolinium MRI · contrast allergy → USG/contrast USG or MRI

Fat → AML · complex cyst → Bosniak · solid enhancing → management options ± renal mass biopsy

Bosniak

Full v2019 table & management → *Ch 136*. Malignancy: IIF 3–10% · III ~50% · IV 75–90%.

- Key discriminator: unequivocal **measurable enhancement** (absent in II/IIF; present in III; enhancing soft tissue in IV).
- Avoid thermal ablation and biopsy in cystic masses — poor yield and risk of tumour spillage. Cystic RCC is indolent, so surgery for III/IV is no longer mandatory; AS is an option.

Renal mass biopsy (RMB)

- Utility-based: obtain when the result may change management, and when lymphoma, metastasis, infection or inflammation is suspected. Not needed in young fit patients heading for surgery regardless, or frail patients managed conservatively regardless. **Multiple cores preferred over fine-needle aspiration.**
- Safe: haematoma ~5%, transfusion <0.5%, seeding almost unreported. Specificity 94%, positive predictive value 98.8%; **non-diagnostic ~14%** (reduced by repeat); subtype accuracy ≥90%, grade 60–80%. A non-malignant result does not prove benignity.

Screening

- No population screening (low yield; most small masses indolent). Not in autosomal dominant polycystic kidney disease.
- ESRD (acquired cystic disease in 80%; RCC in 1–2% of these): USG/CT from 3rd year of dialysis if fit. Continue native-kidney imaging after transplant.
- VHL protocol: eye/physical exam yearly from infancy; metanephrines from 5; CNS MRI + audiogram 2-yearly from 11; **abdominal MRI 2-yearly from 15.**
- Genetic counselling: RCC at **≤46 y**, multifocal/bilateral, suggestive personal/family history or pathology.

6 · STAGING & PROGNOSIS

tumour-node-metastasis (TNM) (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%

tumour-node-metastasis (TNM) (AJCC 8th)	Definition	5-y survival
T3a	Renal vein/segmental branches, pelvicalyceal system , perirenal/sinus fat, within Gerota's fascia	45-75%
T3b	IVC below diaphragm	40-55%
T3c	IVC above diaphragm or caval wall invasion	20-45%
T4	Beyond Gerota's, incl. contiguous ipsilateral adrenal	0-35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0-30%

Stage I T1N0 · II T2N0 · III T1-2N1 or T3 any N · IV T4 or M1.

- **Staging work-up:** abdominal CT + chest X-ray (chest CT if symptoms, abnormal X-ray or high risk); bone scan only with ↑ alkaline phosphatase, bone pain or poor performance status; no PET. Nodes ≥2 cm almost always malignant; smaller nodes often reactive and should not preclude surgery. CT sensitivity: renal vein thrombus 78%, IVC 96% (misses are right-sided). Abnormal/indistinct adrenal → treat as involved; normal → leave it.
- **Pathological stage** = single most important factor; then grade, size, subtype, necrosis, sarcomatoid. Big drop with T4, N1, M1 (node-positive 5-y 5-30%).
- Venous thrombus with otherwise confined disease: 5-y 45-69% — high stage, still curable.
- Adverse clinical factors: symptoms, weight loss >10%, poor performance status, anaemia, hypercalcaemia, ↑ ESR/lactate dehydrogenase (LDH)/alkaline phosphatase, thrombocytosis.
- Integrated nomograms (e.g. Kattan) outperform stage but need revalidation. ASSURE post hoc: recurrence algorithms barely beat stage + grade → AUA risk groups use stage/grade. † **UISS** = TNM + grade + ECOG performance status; **SSIGN** (Mayo) = Stage, Size, Grade, Necrosis; **Leibovich** = pT, pN, size, grade, necrosis → recurrence risk in clear cell RCC.

7 · MANAGEMENT OF LOCALISED RCC

American Urological Association (AUA) 2021 principles

- **Work-up:** multiphase imaging, metabolic panel, blood count, urinalysis, chest imaging; assign **CKD stage (eGFR + proteinuria)** — both independently predict functional decline and survival. Refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30.
- **Counsel** on tumour biology (many cT1a are low risk), morbidity of each option, age, comorbidity, frailty, life expectancy and CKD/dialysis risk.
- **PN prioritised for cT1a** when intervention is indicated; nephron-sparing prioritised for solitary kidney, bilateral or familial tumours, CKD/proteinuria; consider in the young, multifocal disease, or comorbidity threatening future function.
- **RN preferred only if all 3:** (1) high complexity, PN hard even for experts; (2) no CKD/proteinuria; (3) normal contralateral kidney with new baseline GFR likely >45. Otherwise PN, unless overriding safety/oncological concerns.
- Thermal ablation (TA): alternative for cT1a <3 cm; percutaneous preferred; RMB before (preferred) or at TA.
- LND for clinically positive nodes; adrenalectomy only if imaging/operative suspicion; minimally invasive where oncology is not compromised. Examine adjacent parenchyma for medical renal disease; refer to oncology for metastasis or incomplete resection.

Renal function: why nephron-sparing matters

- ~25% of "ideal" PN candidates already have CKD (GFR <60); CKD after RN 65% vs PN 20%.
- **Surgical CKD** (one-time nephron loss) is stable (~0.7%/y) with near-normal survival if GFR >45; **medical CKD** (hypertension, diabetes) progresses (~4.7%/y) and shortens life. Hence the RN rule hinges on a healthy contralateral kidney and GFR >45.
- PN preserves ~90% of global function, RN 65-70%; compensatory hypertrophy 20-30%.

- **Predicted GFR after RN** $\approx$ **global GFR $\times$ contralateral split function $\times$ 1.25** (split function by volumetry).

Active surveillance

- Small masses grow 0.1–0.3 cm/y; metastasis <2%. **DISSRM** registry supports AS; delayed intervention usually still allows nephron-sparing.
- Favouring AS: elderly, life expectancy <5 y, comorbid, frail, marginal renal function; tumour <3 cm, growth <5 mm/y, cystic, non-infiltrative, low complexity. Strong consideration for solid <2 cm or predominantly cystic.
- Imaging every 3–6 mo (AS, with intent to treat) or 6–12 mo (expectant management, often palliative). Repeat at 3–6 mo, consider RMB, then by growth rate.
- **Triggers to intervene:** size >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms. Caution: a subgroup grows fast and delay may lose the PN/TA window.

Partial nephrectomy

Indication	Situations
Imperative	Bilateral RCC, solitary functioning kidney — RN would mean dialysis
Relative	Contralateral kidney threatened (renal artery stenosis, stones, diabetes, hypertension, hydronephrosis)
Elective	cT1a with normal contralateral kidney (standard of care)

- **Pre-op:** exclude local extension/metastasis; 3-D contrast CT (MRI in severe CKD) to map vessels and collecting system.
- **Steps:** temporary pedicle occlusion $\rightarrow$ excision with a rim of normal tissue $\rightarrow$ collecting-system closure and vessel ligation $\rightarrow$ capsular reconstruction $\rightarrow$ *Ch 142*. Negative margin is the priority; its width is immaterial (critical for hilar tumours).
- **Function determinants:** pre-op parenchymal quality (fixed ceiling) $\rightarrow$ **quantity of vascularised parenchyma preserved (primary, modifiable)** $\rightarrow$ ischaemia (secondary if limited). Use hypothermia if ischaemia $\geq$ 25–30 min, especially solitary kidney/CKD. Off-clamp shows no functional gain over limited warm ischaemia.
- **Complications:** urine leak and bleeding commoner than after RN (vascular reconstructed organ); most managed conservatively.
- **Solitary kidney:** temporary dialysis $\sim$ 3.5%, ESRD $\sim$ 4.5%; need $\geq$ 20–30% of one kidney. Loss of >50% parenchyma $\rightarrow$ hyperfiltration, proteinuria, focal segmental glomerulosclerosis $\rightarrow$ angiotensin-converting enzyme (ACE) inhibitor + low-protein diet. Local recurrence 3–5%.
- **Bilateral:** staged; PN side first, then RN if needed. If PN impossible for an imperative indication: RN + dialysis (transplant after cancer-free interval), or neoadjuvant TKI (stop several half-lives before and after surgery — impairs healing).
- **Enucleation** (blunt dissection along pseudocapsule): less bleeding, maximal parenchyma; established in familial RCC; controversial in sporadic, as 40–50% infiltrate the pseudocapsule.

Radical nephrectomy

- Robson principles: early artery then vein ligation, removal outside Gerota's fascia, adrenalectomy, extended LND. Today perifascial dissection remains essential ($\sim$ 25% of cT1b/T2 invade fat), but **no routine adrenalectomy** (unless imaging enlargement or extensive/locally advanced tumour) and **no routine extended LND** (haematogenous spread, variable drainage, no proven benefit).
- Laparoscopic RN standard without gross invasion or major venous involvement; approaches $\rightarrow$ *Ch 141*.

Thermal ablation $\rightarrow$ Ch 143

- Best <2.5–3 cm; for elderly/comorbid, recurrence after PN, hereditary multifocal RCC. Higher local recurrence than PN (cryoablation 3–10%, radiofrequency 5–20%), mostly salvaged by repeat TA; no

difference in metastasis or cancer-specific survival. Central/nodular enhancement in the ablation zone = recurrence → biopsy/retreat. Salvage surgery hampered by fibrosis.

Familial RCC — the 3-cm rule (NCI)

- VHL solid tumours <3 cm: no metastases at ~5 y; ~25% metastasise when larger → intervene at **3 cm** with PN/enucleation/TA to preserve nephrons and delay dialysis. Applies to VHL, hereditary papillary, BHD. **HLRCC is the exception** — FH-deficient tumours metastasise early even when small, so excise at any size with wide margins.
- Recurrence is near-universal (occult microscopic disease) → lifelong surveillance; repeat PN is difficult, so TA helps. Belzutifan cuts interventions (renal response 30–50%; anaemia, fatigue).
- Before renal surgery in VHL, **exclude pheochromocytoma** and CNS haemangioblastoma. Transplant after removal of all renal tissue is safe.

8 · LOCALLY ADVANCED RCC

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Suspect with leg oedema, right-sided/non-collapsing varicocele, dilated abdominal veins, proteinuria, pulmonary embolism, right atrial mass or non-functioning kidney.

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Sequential clamp: infrarenal IVC → contralateral renal vein → suprarenal IVC; occlude lumbar; extract in bloodless field
III	Intrahepatic, below diaphragm	T3b	Mobilise liver, milk thrombus below hepatic veins; or suprahepatic clamp + Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass ± hypothermic circulatory arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins, graft/reconstruct

- **Pre-op:** MRI or CT as close to surgery as possible (thrombus grows); enhancement distinguishes tumour from bland thrombus. **No routine renal artery embolisation** — no benefit, more complications and deaths. Coronary angiography before planned bypass (levels III–IV). Intra-op transoesophageal echo defines the tip, detects emboli, confirms clearance.
- **Sequence:** early arterial ligation, mobilise kidney until attached only by the vein, then thrombectomy (or handle thrombus first to reduce embolism).
- Completely occluded IVC may not need reconstruction (collaterals). Distal bland thrombus → remove, or ligate IVC above it/filter to prevent embolism.
- Complications: bypass coagulopathy, stroke, myocardial infarction; level IV mortality 5–10%.

Locally invasive (T4)

- Only ~40% of suspected invasion is confirmed — large tumours metastasise more readily than they invade (e.g. upper-pole "liver invasion" with intact planes).
- En bloc resection (bowel, spleen, pancreas, abdominal wall) is the only cure; prepare with bowel preparation, splenectomy vaccination, steroid cover if both adrenals at risk. Prognosis still poor. Debulking rarely indicated (1-y survival 10–20%); neoadjuvant therapy as a "litmus test" for borderline resectability.

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant. Node risk factors: tumour ≥10 cm, cT3/T4, grade 3–4, sarcomatoid, necrosis; ≥2 factors → >40% node-positive in a prospective series. 45% of positive nodes lie away from the hilum, so hilar sampling under-stages.

- No proven survival benefit (EORTC 30881) → LND for clinically suspicious nodes, mainly for staging; bulky nodes behave like metastatic disease but resect if feasible.

9 · RECURRENCE, ADJUVANT THERAPY & FOLLOW-UP

Local recurrence	After RN	After PN	After TA
Rate	2-4%	1.4-10%	Cryo 3-10%, radiofrequency ablation (RFA) 5-20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20-40% isolated → full metastatic work-up; en bloc resection → 30-50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- Positive PN margin: linked to recurrence only in high-grade/high-stage disease → usually surveillance; completion nephrectomy only for gross involvement with a healthy contralateral kidney.
- Rationale for adjuvant therapy: 20-35% develop distant metastasis after "curative" surgery (occult micrometastases). Failed: interleukin-2, interferon, medroxyprogesterone, radiotherapy, vaccines, girentuximab, most TKIs. **Adjuvant pembrolizumab** is FDA-approved for high-risk RCC.
- Neoadjuvant TKI: 10-30% shrinkage; may enable PN or thrombectomy; ~20% serious events (wound healing). Metastatic therapy → Ch 144.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1-2	X-ray
Intermediate	pT1 grade 3-4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging from 3-6 mo, intensity rising with risk group; very high risk imaged most often and up to 10 y. Microscopic positive margin → move up one level.
- Every visit: history, examination, creatinine, eGFR, urinalysis for proteinuria → nephrology if progressive. Contrast CT/MRI preferred; USG may alternate after 2 y in low/intermediate risk. **30% of recurrences occur after 5 y** → shared decision beyond 5 y.
- Benign pathology → no routine imaging. After TA: contrast imaging within 6 mo, then intermediate-risk schedule.

10 · OTHER RENAL MALIGNANCIES

Tumour	Clues	Management
Sarcoma (1-2%)	Leiomyosarcoma commonest (50-60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins — first resection is the best chance; margin + grade = prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%) commonest; massive/atypical adenopathy, splenomegaly; prior lymphoma → recurrence until proven otherwise	Biopsy, avoid nephrectomy ; R-CHOP ± radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung commonest; multifocal, 5-30 HU	Biopsy; systemic/palliative; nephrectomy only for refractory bleeding
Carcinoid	Horseshoe kidney (82×)	Excision; good prognosis

Tumour	Clues	Management
Small cell	Usually advanced	Surgery + platinum
PNET	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than children

KEY POINT Almost all non-RCC malignancies are **hypovascular**; the key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis — biopsy, don't resect). RCC nodes land interaortocaval (right) or para-aortic (left); nodes centred elsewhere suggest lymphoma. A solitary enhancing mass years after organ-confined non-renal cancer is more likely primary RCC.

11 • LANDMARK TRIALS

Trial	Design	Result	Take-home
EORTC 30904	RCT RN vs elective PN, ≤5 cm, normal contralateral	PN better function; 10-y overall survival better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT RN ± LND	No survival benefit	Underpowered, low-risk (4% node+); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival 6.8 vs 5.6 y; no overall survival gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	Disease-free survival HR 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

12 • MCQ ONE-LINERS

- Most accepted environmental risk factor for RCC: tobacco.
- VHL gene 3p25; with elongin B/C targets HIF- α for degradation.
- Main HIF isoform in clear cell RCC: HIF-2 α ; inhibitor belzutifan.
- VHL type 2C: pheochromocytoma only; type 1: no pheochromocytoma.
- Only oncogene familial RCC: MET (hereditary papillary, 7q31).
- BHD: FLCN 17p11.2 — chromophobe/hybrid oncocytic tumours, pneumothorax.
- HLRCC: FH 1q42; leiomyomas; excise at any size.
- Papillary RCC: trisomy 7 and 17; clear cell: 3p loss, 5q gain.
- Cup-like CA-IX, CK7+: clear cell papillary tumour (benign).
- Medullary carcinoma: sickle cell trait, SMARCB1 loss.
- Commonest paediatric RCC: Xp11 TFE3 translocation.
- ISUP grade 4: sarcomatoid or rhabdoid differentiation.
- Commonest paraneoplastic finding: raised ESR.
- Stauffer syndrome: interleukin-6; persistence after nephrectomy = residual tumour.
- CT fat < -20 HU = AML; enhancement >15-20 HU = RCC-suspicious.
- TNM 8th: pelvicalyceal invasion = T3a.
- Caval wall invasion = T3c at any level.
- VHL abdominal MRI from age 15, every 2 years.
- Remnant ≥20-30% of one kidney avoids ESRD.
- Hypothermia for PN if ischaemia ≥25-30 min.
- 30% of recurrences occur after 5 years.
- Commonest renal sarcoma: leiomyosarcoma.
- Commonest malignant renal neoplasm overall: metastasis (lung primary).
- Renal carcinoid: horseshoe kidney, 82-fold risk.

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

Reading time ≈ 28 min (slow pace)

1 · EPIDEMIOLOGY

- Upper tract urothelial carcinoma (UTUC) = **5-10%** of urothelial cancers; incidence rising (ureteric tumours).
- Peak age 70-90; male:female ≈ **2:1**. In the Balkans UTUC is **40%** of renal neoplasms.
- Bilateral ≤ 5%; multifocal **10-20%**; concomitant carcinoma in situ (CIS) 11-36%; concurrent bladder cancer **17%**.
- At diagnosis ≈ **60% are invasive** and 7-9% metastatic.
- After UTUC treatment, the bladder recurs in **22-47%** and the contralateral upper tract in 2-6%.

KEY POINT Why UTUC presents invasive: the upper tract urothelium looks like bladder urothelium but sits on a much thinner (mesodermal) muscle layer, so tumours reach muscle, renal parenchyma or periureteric fat earlier than in the bladder.

2 · RISK FACTORS

- Smoking** - the most important modifiable factor; relative risk (RR) **2.5-7**, dose-dependent, ureter > pelvis. Stopping for > 10 yr cuts the risk by 60-70%.
- Occupational "amino tumours"** - benzidine and β-naphthylamine (dyes, rubber, textiles, petrochemicals, coal); mean exposure ≈ 7 yr, latency up to 20 yr.
- Chronic infection, stones, obstruction** - RR 1.5-2; linked to squamous cell carcinoma (SCC), less often adenocarcinoma.
- Iatrogenic** - cyclophosphamide (via its metabolite acrolein), anthranoid laxatives used > 1 yr, pelvic radiotherapy.

Feature	Aristolochic acid (Balkan endemic nephropathy [BEN], Chinese herb nephropathy)	Arsenic (blackfoot disease)	Analgesic abuse (phenacetin)	Lynch syndrome (LS)
Setting	Rural Balkans; Belgium 1992-93 (slimming herbs); Taiwan/China	Well water in SW Taiwan and Chile	Chronic analgesic users	Families with a mismatch repair (MMR) mutation
Mechanism	Aristolactam-DNA adducts → A:T→T:A transversion (TP53 codon 139)	Arsenic; causal link unproven for UTUC	Papillary necrosis; capillarosclerosis (pathognomonic)	MMR deficiency (MSH2 carries the highest risk)
Sex / age	No sex bias; ≈ 10 yr earlier than sporadic	Female 2:1; age 55-60	Latency ≈ 22 yr	< 60 (mean 55); relatively more women
Site	Bilateral more often	Ureter 2:1 over pelvis	Pelvis > ureter; look for contralateral disease	Upper tract > bladder
Inherited?	Familial, not inherited (shared diet)	No	No	Yes, autosomal dominant

KEY POINT All four behave as a **field disease**, with more multiple and bilateral recurrences. That matters for treatment: kidney-sparing options become more attractive, but BEN patients often also have renal insufficiency. An analgesic abuse history is present in 22% of pelvic and 11% of ureteric tumours.

Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC)

- Caused by mutations in the MMR genes MLH1, MSH2, MSH6 and PMS2. UTUC risk is $\approx$ **14x** that of the general population and $\approx$ **75x** in MSH2 carriers. A germline mutation is found in **5-9%** of screened UTUC patients.
- **Screen at the first interview** (European Association of Urology, EAU) if any of: age < 60; a personal history of an LS-spectrum cancer; a first-degree relative (FDR) < 50 with an LS cancer; two FDRs with an LS cancer.
- Suspected cases $\rightarrow$ germline DNA sequencing $\rightarrow$ surveillance for other LS cancers (colorectal, endometrial) $\rightarrow$ family counselling. A sporadic-looking tumour with loss of MMR proteins on immunohistochemistry (IHC) also goes to germline testing.

3 · PATHOLOGY & MOLECULAR BIOLOGY

- > 90% of tumours are urothelial carcinoma (UC). Variant histology occurs in $\approx$ **25%** (squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial) and marks high-grade, aggressive disease. **Micropapillary** is an independent predictor of worse outcome.
- Typical sequence: hyperplasia $\rightarrow$ dysplasia $\rightarrow$ CIS. CIS is easily missed (whitish plaque or red patch).
- Ureteric inverted papillomas harbour malignancy in $\approx$ **18%**, so follow them for $\geq$ 2 yr.
- SCC is 0.7–7% of tumours, 6x commoner in the pelvis, and invasive at presentation. Adenocarcinoma is < 1%, linked to stones and obstruction, with a poor prognosis.
- Ureteric tumours are **70% distal**, 25% mid and 5% proximal, reflecting downstream implantation. Sessile tumours invade in > 80%.
- **Molecular** – UTUC and bladder cancer are genetically distinct but usually clonally related in the same patient. Of five molecular subtypes, **RAS-mutated** has the worst prognosis. Microsatellite instability (MSI) and hypermethylation are the key differences from bladder cancer; MSI is independently prognostic and flags LS. No tissue marker is in routine use.

4 · CLINICAL FEATURES & DIAGNOSIS

- **Haematuria** is the commonest sign (56–98%). Flank pain comes next (20–32%): dull from gradual obstruction, or colicky as clots pass. A mass is found in $\approx$ 10% and $\approx$ 15% are incidental. Systemic symptoms $\rightarrow$ metastatic work-up.
- **Cystoscopy is always done** because bladder cancer often coexists. Positive cytology with normal cystoscopy (including the urethra) $\rightarrow$ evaluate the upper tracts.

Test	Role and key points
Computed tomography urography (CTU) – first line; the EAU prefers it to MR urography	Sensitivity $\approx$ 92% , specificity $\approx$ 95%. Signs: filling defect (50–75%), obstruction, non-visualisation. Misses flat lesions and invasion of small tumours. UC $\approx$ 46 Hounsfield units (HU) vs $\approx$ 100 HU for a uric acid stone.
Hydronephrosis	Means an invasive ureteric tumour in up to 80%; an independent adverse predictor.
Magnetic resonance (MR) urography	Used when CT contrast or radiation is contraindicated. Sensitivity $\approx$ 75% for tumours < 2 cm. Avoid gadolinium if creatinine clearance < 30 mL/min (risk of nephrogenic systemic fibrosis).
Retrograde pyelography	Accuracy $\approx$ 75%.
Fluorodeoxyglucose positron emission tomography/CT (FDG-PET/CT)	Nodes/metastases when iodinated contrast is contraindicated.
Split-function renal scan	Always assess contralateral function before radical surgery.
Flexible ureteroscopy (URS) + biopsy	Raises diagnostic accuracy from 75–85% to $\approx$ 90%. Biopsy gives the grade in $\approx$ 90%; visual grading is right in only $\approx$ 70%.

Test	Role and key points
Urine cytology	Sensitivity $\approx$ 65%, depending on grade: G1 20%, G2 45%, G3 75%. Selective ureteric washing beats voided urine; brush biopsy + cytology reaches $\approx$ 90%. Collect before contrast , which degrades the cells.
Fluorescence in situ hybridisation (FISH)	More sensitive than cytology; best for low-grade surveillance after kidney-sparing treatment; not routine.

KEY POINT Why URS biopsy cannot stage: samples are small and rarely include deep layers, so a Ta biopsy is often upstaged to T3 at resection. Stage is therefore **inferred** from grade, appearance, hydronephrosis and cytology, and patients treated with kidney-sparing surgery need intensive follow-up.

- **Indications for diagnostic URS:** diagnostic uncertainty; kidney-sparing treatment under consideration; solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence, especially with pelvic tumours and biopsy, although survival is unchanged. Avoid it when CTU and cytology already justify RNU.

5 • STAGING & GRADING

Tumour, node, metastasis (TNM, 8th ed.)	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or renal parenchyma ; ureter: periureteric fat. pT3a = microscopic parenchymal invasion; pT3b = macroscopic invasion or peripelvic fat (worse)
T4	Adjacent organs, or through the kidney into perinephric fat
N1 / N2	Single node $\leq$ 2 cm / single node $>$ 2 cm or multiple nodes

- American Joint Committee on Cancer (AJCC) stage: 0a/0is; I = T1; II = T2; III = T3; **IV = T4, any N+ or M1.**
- Nodes: pelvis/upper ureter $\rightarrow$ hilar and retroperitoneal; lower ureter $\rightarrow$ iliac/pelvic.
- Grade by World Health Organization (WHO) 2004/2016: papillary urothelial neoplasm of low malignant potential (PUNLMP), low grade, high grade. PUNLMP is managed as low grade.

6 • PROGNOSTIC FACTORS & RISK STRATIFICATION

- **Stage is the single most important factor**, then grade. 5-yr cancer-specific survival (CSS) in high-risk RNU cases (Surveillance, Epidemiology and End Results [SEER] data): $\leq$ T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse than pelvic tumours** at the same stage. The renal parenchyma acts as a barrier, whereas periureteric tissue opens straight into a rich vascular and lymphatic plexus.
- **Preoperative factors:** age (predictive, but never a reason by itself to deny curative RNU); smoking; high American Society of Anesthesiologists (ASA) score and poor Eastern Cooperative Oncology Group performance status (ECOG-PS); obesity; adverse blood markers (low albumin, high neutrophil-to-lymphocyte ratio, high C-reactive protein); **surgical delay - do RNU within 12 wk** for invasive disease.
- Size: **2 cm** is the best cut-off for predicting $\geq$ pT2.

- **Postoperative factors:** nodal metastasis (up to 40%) and extranodal extension; lymph node density $\geq 30\%$; lymphovascular invasion (LVI, $\approx 20\%$; independent in node-negative disease and never seen on biopsy, so it must be reported on every specimen); positive margins; necrosis $> 10\%$; sessile architecture; concomitant CIS; variant histology.

Low risk — ALL required	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
< 2 cm	> 2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy
—	Variant histology

KEY POINT Because true stage is known only after RNU, every preoperative factor is a stand-in for invasion. The rule is asymmetric: a single adverse feature makes the tumour high risk. **Trap:** a small, low-grade unifocal tumour **with hydronephrosis is high risk.**

7 • RECURRENCE PATTERNS

- **Bladder after UTUC:** 15–75% at 5 yr, usually within 2 yr.
- **Upper tract after bladder cancer:** 2–4%, late (median > 3 yr); 0.75–6.4% after cystectomy. About **30%** of patients with bladder CIS treated with bacillus Calmette-Guérin (BCG), or cystectomised for BCG-refractory disease, develop UTUC. Risk factors: distal ureter/orifice tumour, multifocal or recurrent non-muscle-invasive disease, CIS, reflux.

KEY POINT Why recurrence is asymmetric: urine flows downhill. Tumour cells seed antegrade, so bladder recurrences are common, early and monoclonal. The field effect (independent clones in carcinogen-exposed urothelium) explains multifocal tumours of different clonal origin; both operate.

Patient factors	Tumour factors	Treatment factors
Male, prior bladder cancer, smoking, chronic kidney disease (CKD)	Positive cytology, ureteric location, multifocal, invasive pT stage, necrosis	Laparoscopic approach, extravesical cuff removal, positive margins, diagnostic URS

Predictors of intravesical recurrence after RNU

- **Pan-urothelial disease** = bladder plus ≥ 2 extravesical sites (women: both upper tracts; men: upper tract(s) and/or the prostatic urethra). There is no evidence-based algorithm. Cystectomy is used for high-grade bladder disease, conservative treatment for non-infiltrating UTUC, and total urinary tract removal is an option in younger patients. Cystectomy followed by RNU loses renal function and can make the patient ineligible for cisplatin.

8 • MANAGEMENT OVERVIEW & FOLLOW-UP

ALGORITHM

Cystoscopy + CTU $\pm$ URS biopsy/cytology + contralateral function $\rightarrow$ LS screening if indicated $\rightarrow$ risk-stratify

Low risk $\rightarrow$ kidney-sparing treatment (endoscopic or segmental) with intensive surveillance

High risk $\rightarrow$ RNU with bladder cuff $\pm$ lymph node dissection (LND), within 12 wk

Clinically node-positive (cN+) $\rightarrow$ systemic chemotherapy first $\rightarrow$ surgery if there is a response

After RNU $\rightarrow$ single postoperative intravesical chemotherapy dose $\rightarrow$ cystoscopic follow-up for everyone

- Preoperative nomograms predict $\geq$ pT2 or non-organ-confined disease and are used to select patients for neoadjuvant chemotherapy; pathological downstaging improves overall survival. Perioperative chemotherapy (neoadjuvant, POUT) $\rightarrow$ *Ch 139*.
- **Why give chemotherapy before surgery?** RNU removes a kidney, and the resulting loss of renal function often makes patients ineligible for cisplatin afterwards.
- LND at RNU gives the best staging; whether it cures is debated.
- A single postoperative intravesical instillation reduces bladder recurrence after RNU.
- Surgical technique (RNU steps, distal ureter management, endoscopic ablation, segmental ureterectomy, topical instillation) $\rightarrow$ *Ch 139*.
- **Follow-up:** cystoscopy for all patients. After kidney-sparing treatment, follow intensively with URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory disease), check the upper tracts every year with CT, which should replace cytology and excretory urography. The risk stays constant, so surveillance is prolonged.

9 • MCQ ONE-LINERS

- | | |
|---|---|
| 1. UTUC = 5-10% of urothelial cancers; Balkans: 40% of renal neoplasms. | 9. Ureteric inverted papilloma: malignancy $\approx$ 18%; follow $\geq$ 2 yr. |
| 2. Smoking RR 2.5-7; ureter at greater risk than pelvis. | 10. CTU: sensitivity 92%, specificity 95%. |
| 3. Lynch: UTUC risk 14 $\times$; MSH2 carriers 75 $\times$. | 11. UC $\approx$ 46 HU; uric acid stone $\approx$ 100 HU. |
| 4. RAS-mutated molecular subtype has the worst prognosis. | 12. Cytology sensitivity: G1 20%, G2 45%, G3 75%. |
| 5. Aristolochic acid signature: A:T $\rightarrow$ T:A transversion, TP53 codon 139. | 13. Pelvic tumour invading renal parenchyma = T3, not T4. |
| 6. BEN: familial, not inherited; bladder cancer not increased in affected families. | 14. N1 = single node $\leq$ 2 cm; any N+ = AJCC stage IV. |
| 7. Arsenic UTUC: blackfoot disease; women 2:1. | 15. Necrosis > 10% is an independent adverse predictor. |
| 8. Phenacetin latency $\approx$ 22 yr; capillarosclerosis is pathognomonic. | 16. Lymph node density $\geq$ 30% predicts poor outcome. |
| | 17. Uroplakin III and p27 expression are favourable. |

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Reading time ≈ 33 min (slow pace)

Epidemiology, risk factors, pathology, staging and imaging → Ch 138

1 · PRINCIPLES & PATIENT SELECTION

- **Standard therapy** for upper tract urothelial carcinoma (UTUC): radical nephroureterectomy (RNU) with bladder cuff, provided the contralateral kidney is normal and the patient is fit.
- UTUC is **unforgiving of spillage**: it implants wherever tumour-laden urine goes, so every approach keeps the collecting system closed.
- **Work-up**: computed tomography (CT) or magnetic resonance imaging (MRI) urogram → cystoscopy → ureteroscopy (URS) with biopsy and selective cytology. URS raises diagnostic accuracy from ≈75% (urography) to 85–90% and is essential when diagnosis is uncertain or kidney-sparing management (KSM) is planned.
- URS biopsies are tiny and shallow, so they cannot stage. Stage is **inferred** from grade, endoscopic appearance, imaging and cytology; low grade (LG) tumours are mostly Ta/T1, high grade (HG) mostly invasive.

American Urological Association / Society of Urologic Oncology (AUA/SUO) risk group	Features	Ablation	Systemic therapy
Low - favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	Preferred	No
Low - unfavourable	LG + obstruction, multifocality or prior lower-tract tumour	May be offered	No
High - favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High - unfavourable	HG cytology, radiographic invasion, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

KEY POINT Low grade does not automatically mean ablation: obstruction or multifocality moves an LG tumour into the unfavourable group, where RNU retains a role.

ALGORITHM

Imaging suspicion → CT/MR urogram → endoscopy + biopsy + cytology → risk group + shared decision

High risk + adequate glomerular filtration rate (GFR) → RNU ± neoadjuvant chemotherapy · Low risk or poor GFR → kidney-sparing

- **Elective KSM**: low-risk disease, irrespective of the contralateral kidney.
- **Imperative KSM** (selected high risk): chronic kidney disease (CKD), solitary kidney, bilateral tumours, unfit patient.
- **Lower ureteric** tumours, especially unfavourable → distal ureterectomy rather than endoscopy.
- **Expectant management** (AUA/SUO option): only with major comorbidity, competing mortality or high end-stage renal disease (ESRD) risk

2 · ENDOSCOPIC (KIDNEY-SPARING) MANAGEMENT

- Principle as in the bladder: sample, then ablate with cautery or laser. **Grade is the key selector** and the strongest predictor of recurrence; results for HG are poor.

- AUA/SUO: endoscopic treatment for tumours **<1.5 cm**; caution ≥ 1.5 cm (invasion risk).
- **Staged approach** for bulky or suspected HG tumour: biopsy and partial ablation only, so that perforation is avoided if RNU follows; complete treatment once pathology confirms suitability. Unresectable, HG or invasive $\rightarrow$ RNU.
- Versus RNU: similar survival but **$\sim 3\times$ more reinterventions**; lifelong surveillance.

	Retrograde ureteroscopic	Antegrade percutaneous
System	Closed - non-urothelial surfaces not exposed	Open - tract seeding $\approx 0.3\%$, all HG
Best for	Low-volume ureteric and small renal tumours	Large pelvis/upper ureteric tumours, lower-pole calyx, urinary diversion
Staging	Not possible	Possible - base biopsied separately
Morbidity	Low	Higher ($\approx 25\%$), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical Rx	Repeat URS; catheter or stent reflux	Via retained nephrostomy

Ureteroscopy - steps

- Cystoscopy $\rightarrow$ cytology $\rightarrow$ wire or no-touch access $\rightarrow$ 6.9–7.5 Fr semirigid scope $\rightarrow$ flexible scope for all calyces $\rightarrow$ map lesions $\rightarrow$ saline barbotage improves cytology yield.
- Tight ureter: pre-stent rather than forceful dilation.
- Techniques: bulk excision + base ablation (LG on stalk); resection only to lamina propria (thin ureter); **biopsy then laser** (usual).
- Bugbee electrode: unpredictable depth; **never fulgurate circumferentially** (stricture).

Laser (yttrium-aluminium-garnet, YAG)	Penetration	Role
Holmium (Ho:YAG)	<0.5 mm	Ideal for ureter
Neodymium (Nd:YAG)	5–6 mm	Transmural injury risk in ureter
Thulium	<0.2 mm	Strong coagulation

Percutaneous - steps

- Retrograde ureteric catheter $\rightarrow$ contrast $\rightarrow$ puncture. Peripheral calyceal tumour: puncture the calyx distal to it, never through it. Pelvis/upper ureter: upper or middle calyx.
- Dilate to 30 Fr. Cold-cup forceps (LG, thin stalk), resectoscope for broad-based tumours (**bipolar**, avoids hypo-osmotic irrigant absorption) or laser. Base sent separately for staging.
- **Nephrostomy always left** $\rightarrow$ second-look nephroscopy and route for topical therapy. HG or invasive histology $\rightarrow$ RNU.
- Diversion: antegrade wire into conduit/pouch, then retrograde URS over it.

KEY POINT Does URS spread tumour? Prior ureteroscopic manipulation increases bladder recurrence (downstream seeding), but pre-RNU URS has not increased metastasis. Mitigate with a single postoperative intravesical instillation.

3 • SEGMENTAL URETERECTOMY

- Preserves nephrons **and** yields a specimen that guides adjuvant therapy. In selected patients outcome equals RNU: stage and grade, not extent of surgery, set prognosis.
- **Ureteroscopy of the entire tract above the tumour is mandatory** to exclude multifocality. HG $\rightarrow$ add ipsilateral pelvic lymph node dissection (LND).
- Ipsilateral recurrence after conservative ureteric surgery **33–55%**, mostly distal to the original site, and lifelong.

Procedure	Indication	Key steps
Segmental resection + uretero-ureterostomy	LG proximal/mid tumour not ablatable (size, number); HG when unit must be saved. Rare.	Ligate 1-2 cm above and below; frozen-section margins; spatulate; 4-0 polyglactin; stent; tension-free with preserved blood supply
Distal ureterectomy + reimplant	LG low-stage, or selected HG/invasive distal tumour when renal preservation needed	Whole distal ureter + cuff; 2-layer posterior cystotomy closure; $\pm$ LND

- Reimplant: **non-refluxing** confines infection and theoretically prevents upper-tract seeding; **refluxing** eases later upper-tract surveillance. Debated.
- **Boari flap**: preoperative cystogram; small or irradiated bladder contraindicates; base ≥ 2 cm wider than apex (blood supply); width $\geq 3 \times$ ureteric diameter.
- **Ileal ureter**: 20-25 cm ileum ≥ 15 cm from ileocaecal valve, isoperistaltic.
- Post-op: drain, Foley 5-7 d, cystogram before removal after extensive reconstruction.

Manoeuvre	Length gained	Condition
Psoas hitch	≈ 5 cm	Mobile bladder; avoid nerve entrapment
Boari flap + hitch	Further 10-15 cm	Adequate capacity
Renal descensus + nephropexy	Up to 8-10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 • POSITIVE CYTOLOGY & CARCINOMA IN SITU (CIS)

ALGORITHM

Positive voided cytology $\rightarrow$ repeat $\rightarrow$ CT urogram + bladder biopsies $\rightarrow$ bladder positive: treat bladder

Persistent positive with negative bladder (or after bacillus Calmette-Guérin (BCG) "6+1") $\rightarrow$ selective upper-tract cytology + retrograde pyelography (RPG) + prostatic urethral biopsy (men) + URS

- Upper-tract CIS is a diagnosis of exclusion. **RNU for cytology alone is not recommended**: cytology gives false positives (infection, stones) and contralateral disease often follows. Segmental resection fails (field change).
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy. Initial response is good, but late recurrences can be lethal, so follow-up must be vigilant. RNU if more than surface disease is proven.

5 • TOPICAL THERAPY IN THE UPPER TRACT

- Indications: solitary kidney, bilateral UTUC, ESRD risk. Liquid agents drain quickly by gravity, so delivery route determines efficacy.
- Routes: antegrade via nephrostomy (gravity, ≤ 25 cm H₂O column); retrograde ureteric catheter; reflux via indwelling stent.
- **Always low pressure and sterile urine**: high pressure and infection cause sepsis and systemic absorption (deaths reported).
- Adjuvant instillation after papillary ablation has **no proven benefit**; adjuvant BCG acceptable after complete ablation of HG favourable Ta/T1.
- Complications: sepsis, stricture; BCG - renal granulomas (commonest), disproportionate function loss with reflux.
- **UGN-101 (Jelmyto)**: mitomycin C (MMC) in reverse-thermal hydrogel — liquid when instilled, gels at body temperature and dissolves over hours, prolonging contact. US Food and Drug Administration (FDA) approved 2020 for LG UTUC **not endoscopically manageable**. Toxicity: ureteric stricture $\approx 44\%$ (fewer antegrade), marrow suppression, embryo-fetal toxicity.

6 · RADICAL NEPHROURETERECTOMY

Indications

- Non-metastatic high-risk tumour of renal pelvis or proximal ureter (HG biopsy/cytology, obstruction, parenchymal invasion), with bladder cuff and LND.
- LG with unfavourable features (obstruction, multifocal, large volume) or failed/infeasible endoscopy.

Open RNU

- Genitalia in field for catheter access. Midline: best for nodes and bladder, poor upper pole; flank incisions need a second incision for the cuff.
- **Kidney first, cuff last:** line of Toldt → colon medial → **clip ureter early** (stops tumour cells being milked into the bladder) → control hilum before manipulation → artery then vein → kidney outside Gerota's fascia.
- **Adrenalectomy not routine.**
- If ureter divided (clips/ties/stapler), **never seal it with a vessel-sealer alone.**

Distal ureter & bladder cuff

- Remove the **entire intramural ureter and orifice:** a retained stump recurs in **30-75%**, cannot be practically surveyed, and worsens survival. After bowel diversion, excise the ureteroenteric anastomosis with a bowel cuff.
- Principles: excise orifice with small urothelial margin → contain spillage → watertight closure.

Technique	Status
Extravesical (laparoscopic/robotic; open with clamp)	Acceptable. Hazards: entering the tunnel, contralateral orifice
Transvesical (open cystotomy)	Acceptable; 1-cm mucosal cuff
Transvesical "lasso"	Abandoned – spillage, recurrence
Endoscopic "pluck"	Abandoned – extravesical spillage
Intussusception "stripping"	Abandoned – recurrence
Laparoscopic "staple"	Abandoned – urothelium trapped in staple line → positive margins

- **Extravesical steps:** follow ureter under superior vesical artery → tie at detrusor → partly fill bladder → divide lateral pedicle → empty bladder → excise cuff en bloc.
- Minimally invasive: open bladder supralaterally so it drains without spillage; flexible cystoscopy confirms complete excision.
- Closure: two layers absorbable; Foley several days; drain optional.

Laparoscopic & robotic RNU

- Specimen extracted **intact in a bag** via incision that also serves the distal ureterectomy.
- Wide perihilar margin (lymphatic spread).
- Ureteric blood supply: anteromedial (upper ⅓), medial (middle ⅓), **lateral** (lower ⅓).
- Relative contraindications: bulky tumour invading neighbours; extended post-chemotherapy LND.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

KEY POINT Approach matters less than principle: port-site seeding and peritoneal carcinomatosis follow spillage, not the approach.

Lymphadenectomy

- Staging, prognostic and therapeutic value in **T2-T4**, especially renal pelvis/proximal ureter.
- Removing **≥8 nodes** improves detection and disease-specific survival above pT1. Surgeon, not approach, determines yield. Specific complication: chylous leak.

Primary	Right	Left	Caudal limit
Renal pelvis	Hilar, paracaval, retrocaval, interaortocaval	Hilar, para-aortic	Inferior mesenteric artery
Upper ⅔ ureter	As renal pelvis		Aortic bifurcation
Lower ⅓ ureter	Ipsilateral iliac, obturator, presacral		Pelvis

Lymph drains right → left, so right-sided tumours need interaortocaval nodes; on the left they add little.

Results

- Prognostic cliff between **T2 and T3**; HG behaves like T3; node-positive disease with surgery alone behaves almost like metastatic disease — the case for systemic therapy.
- Bladder recurrence (drop-down implantation) is common; loss of renal function is RNU's main cost.

7 • PERIOPERATIVE ADJUNCTS

Single-dose intravesical chemotherapy

- Bladder tumours develop in 30-50% after RNU or endoscopic treatment, mostly within 2 years, from seeding. **AUA/SUO strongly recommend** one dose after RNU (optional after endoscopy).
- Postoperative (3-7 d): cystogram excludes cuff leak → MMC or gemcitabine → dwell 1-2 h.
- **Intraoperative** (authors' preference): instil before surgery, clamp 1-2 h, drain before cuff dissection. Avoids cystogram and leak-related cancellation; drug present when seeding occurs.

Perioperative systemic chemotherapy (high risk)

- **Renal function is the hinge**: CKD is common and worsens after RNU, so many lose cisplatin eligibility — hence give platinum **before** the kidney is removed.
- **Neoadjuvant**: AUA/SUO standard option for high-risk UTUC, especially if post-op creatinine clearance (CrCl) expected **≤50** mL/min. EA8141: CrCl >50 → accelerated MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×4; 30-49 → gemcitabine-carboplatin; pathological complete response 14%, <ypT2N0 60%.
- **Adjuvant** (POUT): pT2-T4 or N+ → gemcitabine-cisplatin (CrCl >50) or -carboplatin (30-49). Less effective with carboplatin and N+ → cisplatin preferred.
- Regional nodes on imaging → chemotherapy first, then consolidative surgery.

8 • COMPLICATIONS & THEIR MANAGEMENT

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; avoid circumferential fulguration
Percutaneous	Bleeding/transfusion, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 • FOLLOW-UP

- Every visit: history, examination, metabolic panel, **cystoscopy**, cytology; after KSM add ipsilateral URS. Do not routinely scope a normal contralateral tract.
- CT/MRI urography; RPG if contrast impossible. Filling defects need URS.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6-9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6-9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3-6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3-6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	+ cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3-6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3-6-monthly ×2 yr; chest CT 6-12-monthly ×5 yr

10 • LANDMARK TRIALS

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% → 16% at 1 yr	Single-dose instillation is standard
Ito (pirarubicin)	RCT, single dose ≤5 d post-RNU; biopsy-proven	42% → 17% at 2 yr	Confirms benefit
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2-4/N+	Stopped early for efficacy; disease-free survival (DFS) hazard ratio 0.45 ; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

11 • MCQ ONE-LINERS

1. Retained intramural stump recurrence: 30-75%.
2. AUA/SUO endoscopic size cut-off: <1.5 cm.
3. Most important predictor of recurrence after endoscopic treatment: grade.
4. Distal-third ureteric blood supply is lateral.
5. Ho:YAG penetration <0.5 mm; Nd:YAG 5-6 mm; thulium <0.2 mm.
6. Percutaneous tract dilated to 30 Fr; tract seeding ≈0.3%, all high grade.
7. Psoas hitch ≈5 cm; Boari flap further 10-15 cm.
8. Boari: width ≥3× ureteric diameter; base ≥2 cm wider than apex.
9. Ileal ureter: 20-25 cm ileum, ≥15 cm from ileocaecal valve.
0. ≥8 nodes removed improves disease-specific survival above pT1.
1. Right renal pelvis template includes interaortocaval nodes.
2. Renal pelvis template caudal limit: inferior mesenteric artery.
3. Bladder recurrence after RNU: 30-50%, mostly within 2 years.
4. UGN-101 (Jelmyto): FDA 2020, low-grade UTUC; stricture ≈44%.
5. Commonest BCG upper-tract complication: renal granulomas.
6. Upper-tract CIS: 6-week BCG induction, preferably via nephrostomy.
7. Antegrade instillation pressure column ≤25 cm H₂O.
8. Neoadjuvant chemotherapy favoured when post-RNU CrCl expected <50.
9. First ureteroscopic tumour treatment: Goodman 1984; antegrade: Tomera 1982.
0. First laparoscopic nephroureterectomy: Clayman 1991.

141 Contemporary Open Surgery of the Kidney

Reading time ≈ 33 min (slow pace)

1 · ROLE & PREOPERATIVE WORK-UP

- Open surgery = **exposure + early great-vessel control**: locally advanced renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours, hostile abdomen. Minimally invasive → Ch 142; RCC staging → Ch 137.
- Renal function**: urinalysis, serum creatinine (SCr), estimated glomerular filtration rate (eGFR) by the **Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) 2021 race-free** equation, more accurate than Modification of Diet in Renal Disease (MDRD). It decides partial (PN) vs radical nephrectomy (RN), counselling and dialysis planning. Cystatin C: independent of muscle mass/diet, detects mild GFR loss.
- GFR **≤60** or proteinuria → nephrology consult.
- Advanced RCC: blood count, metabolic panel (Memorial Sloan Kettering [MSKCC] and International Metastatic RCC Database Consortium [IMDC] risk models), liver tests + coagulation (**Stauffer syndrome** — paraneoplastic hepatic dysfunction), cardiopulmonary assessment, because positioning, blood loss and fluid shifts can decompensate heart and lungs.
- Imaging maps kidney, vessels and neighbours (colon, pleura, pancreas, spleen, liver): computed tomography (CT)/magnetic resonance imaging (MRI); contrast generally safe if GFR **≥30**. Suspected thrombus → MR or CT venography, **≤1 month old** (rapid propagation).
- Prophylaxis: cefazolin (or clindamycin) ≤1 h pre-incision; urine culture only if infection suspected; no routine bowel prep; pneumatic compression + heparin 5000 IU subcutaneous (SC); stop smoking 4–6 wk; spirometry.
- Renal artery embolisation**: authors no longer use routinely. Postinfarction syndrome (pain, fever ≤72 h); may *increase* blood loss (oedema); harder dissection if surgery delayed >48 h. If used, operate same/next morning and confirm inflow is occluded.

2 · SURGICAL ANATOMY & APPROACHES

- At risk: **right** — liver, duodenum, colon; **left** — spleen, pancreatic tail, colon, superior mesenteric artery (SMA).
- Right renal artery**: posterior-inferior to renal vein, behind IVC; reachable in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- SMA**: anterior aorta **1–2 cm above LRV**, veers left → mistaken for left renal artery.
- Right gonadal/adrenal veins → IVC; left → LRV.
- Anterior exposure = colon off Gerota's via **white line of Toldt**; right adds **Kocher** (no cautery — duodenal injury devastating); left divides splenorenal ligament.
- Cattell-Braasch**: Toldt → caecum → Kocher → mesenteric root to Treitz; midgut pivots on SMA, exposing aorta/IVC. **Infracolic approach** (trauma, bulky hilum): incise below Treitz → aorta + LRV.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour/IVC thrombus (authors' choice — early aortic access); bilateral	Bowel mobilisation
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN + thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN/PN; 8-cm mini-flank supra-11th	Pleura

Incision	Best for	Limitations
Transcostal (11th, rib resected)	Retroperitoneum at old rib tip	Rib division
Thoracoabdominal, dorsal lumbotomy	Largely historical (morbidity); thoracoabdominal still an option for level III-IV thrombus	

- **Supine:** lumbar roll/flexion; excess hyperlordosis → back pain (avoid in spinal stenosis); aortocaval compression; pad elbows.
- **Flank:** 12th rib tip over kidney bar, axillary roll; ↓ venous return and lung function — poor for cardiopulmonary disease, severe kyphosis; pad against rhabdomyolysis.
- Higher rib = better upper-pole/hilar exposure but nearer pleura. Spare **subcostal nerve** (between internal oblique and transversus); cut intercostals on **upper border of rib**.

3 • SIMPLE NEPHRECTOMY

- **Within Gerota's fascia** for benign nonfunction (obstruction, infection, stones, reflux, polycystic, dysplasia); functioning kidney rarely (pain, bleeding, renovascular hypertension, infection).
- Stay on, not in, the capsule; include psoas fascia if adherent; aspirate large hydronephrosis; spare adrenal.
- Benign PN: focal atrophy, duplex pole, infected diverticulum, symptomatic angiomyolipoma.

4 • RADICAL NEPHRECTOMY

- Robson: kidney + Gerota's + adrenal + nodes crus → bifurcation. Now: **spare normal adrenal**, selective lymph node dissection (LND).
- **Indications:** not PN-amenable — nonfunctioning kidney, tumour replacing kidney, nodes, cytoreduction, venous thrombus.
- Staging: abdominal CT/MRI + **chest CT** (plain film misses small/mediastinal disease); bone/brain imaging only if symptomatic.
- Biopsy if it changes plan: infiltrative mass (urothelial, lymphoma, metastasis), before systemic therapy, bilateral tumours, trials.
- T4 invasion suggested by altered organ enhancement, not contact. Liver rarely truly invaded.

ALGORITHM

Right: survey abdomen → Toldt, iliac → hepatic flexure (protect right colic/ileocolic) → Kocher → IVC → loop and palpate renal vein → artery behind vein → **artery then vein** → kidney outside Gerota's.

Difficult hilum: interaortocaval artery ligation → kidney decompresses → flaccid vein checked, double-ligated → lateral artery divided. Or: lower pole → divide ureter → lift kidney to show artery.

Left: Toldt splenic flexure → iliac → renocolic ligament (pad pancreas) → aorta leads to LRV; gonadal vein leads up to it → artery above/behind LRV → 0 silk + 3-0 Prolene, confirm occlusion → vein.

KEY POINT Why artery first? Tying the vein first leaves the kidney engorged with arterial inflow and collaterals bleed. Arterial ligation shrinks the kidney and softens the vein so it can be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or that the SMA was tied.

- **En bloc pedicle** (hilar nodes): 3 clamps, divide 2nd-3rd, tie ×2 + suture ligature; arteriovenous fistula (AVF) theoretical, very rare.
- **Lumbar veins:** silk ties; **no clips on IVC side** (they block emergency caval clamping). Retracted lumbar: sponge-stick psoas compression → fulgurate/suture.
- **Loss of pedicle control:** tell anaesthetist, resuscitate, finger/sponge compression, 2 suckers, **never clamp blindly**. Artery: compress aorta above, clamp, 2-layer repair. IVC hole: finger → Allis

→ suture.

- Prolene: IVC/aorta 3-0/4-0; renal vessels 4-0/5-0; BB needle on arteries (spares calcified plaque).
- **Adrenalectomy** only for upper-pole tumour with lost plane, adrenal mass, diffuse involvement. Metastasis **≤5%**; contralateral as common → routine removal risks insufficiency.
- **LND**: nodes positive ≈5%. European Organisation for Research and Treatment of Cancer (EORTC) randomised trial (772 patients): no overall survival (OS) benefit, but many small low-stage tumours enrolled. LND is prognostic, may reduce retroperitoneal recurrence. **Do if nodes enlarged** (short axis **≥1.5 cm ≈ 50% positive**); consider in high risk (cytoreduction, >10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- Templates: right — paracaval, precaval, retrocaval, interaortocaval, crus → bifurcation; left — para/retroaortic, LRV → bifurcation. **Clip lymphatics above renal vein** → no chylous ascites.
- **Closure**: saline in wound + big breath; bubbles = pleural tear → suture over 14-Fr catheter, withdraw on held breath; **chest X-ray in all**.

5 · IVC TUMOUR THROMBECTOMY

- RCC = commonest cause; **5-10%** of RCC. Untreated → most die within 1 yr, so surgery (± systemic therapy) is the backbone, with RN + LND.
- **Tumour thrombus** (contains tumour) vs **bland thrombus** (clot below obstruction) — managed differently.
- Preop: screen for pulmonary embolism (PE)/deep vein thrombosis; anticoagulate if venous thromboembolism, bland thrombus or IVC occlusion. **No suprarenal filters** (thrombus grows onto them). Embolise rarely (unresectable Budd-Chiari, bleeding kidney).
- **Transoesophageal echo (TEE)** for level II-IV. Vascular/liver surgeon II-III; **cardiothoracic for III-IV**, planned.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC ≤2 cm above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At/above major hepatic veins, below diaphragm	Liver mobilisation ± Pringle; occlusive → suprahepatic clamp; nonocclusive → venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) + deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip/staple); don't extract — organised clot fails to clear, vessel injured
D	Mixed with tumour	Resect infrarenal IVC; keep lumbar as collaterals

ALGORITHM

Level I-II: early renal artery ligation → tourniquets **infrarenal IVC** → **contralateral renal vein** → **suprarenal IVC** → cavotomy → extract → heparinised saline flush, inspect wall (biopsy/resect suspicious intima) → 4-0 Prolene → release **caudal clamp first**, flush 5-10 mL on held breath → cranial clamp.

- **Left tumour:** harder — IVC reached from the right, both colons mobilised. Authors don't clamp right renal artery (no venous collaterals); intermittently release right renal vein clamp. No routine

heparin — bleeding, not clotting, is the problem.

- **Level III:** bypass on standby; transplant-style liver mobilisation. **Major hepatic veins never divided**; accessory (caudate) veins tied → 2–3 cm more IVC. **Budd-Chiari = contraindication.**
- **Pringle** needed if clamping above hepatic veins (else liver engorges and bleeds): ≤60 min, ideally **≤20 min** (liver ischaemia, portal thrombosis, splenic rupture).
- **Clamp trial** 2–5 min: preload ↓60%, cardiac output (CO) ↓40%. **CO ↓>50% or mean arterial pressure ↓>30%** → bypass. Proceed if tolerated and extraction <30 min. Occlusive thrombus tolerates clamping (azygos collaterals); free-floating does not.
- Bleeding and hypotension cause incomplete tumour removal, which is universally fatal, so never compromise IVC control to save time.
- Also bypass for cardiac/hepatic dysfunction, poor contralateral kidney, portal hypertension, uncontrolled bleeding.

	VVB	CPB + DHCA
Access	Femoral → jugular, percutaneous	Sternotomy; 16–18 °C, drain 95% blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar/intercostal back-bleeding	Cardiac, neurological
Use	Least invasive; not with atrial or iliac bland thrombus, Budd-Chiari	Level IV; bloodless field; ≤60 min

- **Reconstruction:** narrowing to 50% → primary closure; more → **patch** (bovine pericardium, smooth side in); width = normal diameter × fraction resected × 3 (Leibovich-Bower). Circumferential → **polytetrafluoroethylene (PTFE) 16–20 mm** graft, wrapped in omentum; low-dose heparin then lifelong aspirin 81 mg.
- Suprarenal IVC ligation: right tumour — OK if LRV tied caval to its tributaries (collateral drainage). Left tumour — right kidney has no collaterals → needs alternative drainage.

KEY POINT "Shortcuts" that become "longcuts": opening IVC before all inflow controlled; side-biting a level I; clips on IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping test clamp.

6 · PARTIAL NEPHRECTOMY

- **Treatment of choice for T1** even with normal function; size-independent if safe. Old imperative indications: solitary kidney, bilateral tumours, renal failure risk.
- **Relative contraindications:** <20% nephron mass left; pedicle encasement, collecting-system invasion, venous thrombus, cT4, cN1.
- **Hyperfiltration:** fewer nephrons each filter more → focal segmental glomerulosclerosis, proteinuria, chronic kidney disease (CKD); rare unless nephron mass ↓>80%.
- Multifocality: clear cell ≈2%, papillary ≈10% → inspect whole kidney. Hereditary: observe to **3 cm**, except **HLRCC** (hereditary leiomyomatosis and RCC) → early wide excision; use intraoperative ultrasound (IOUS).

Question	Evidence
Warm ischaemia	Aim ≤30 min ; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss, stay

KEY POINT PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes, so ischaemia is individualised; hypothermia and mannitol are optional (role in solitary kidney/CKD unclear).

- Always **control hilum first**, even if not clamping, so a clamp can go on at once. Options: manual, parenchymal clamp, selective branch, whole pedicle.
- IOUS for endophytic/central tumours, before clamping.
- **Enucleation**: pseudocapsule plane, minimal normal tissue. **Resection**: 5–10 mm rim, usually clamped. Frozen section of deep margin if needed.
- **Renorrhaphy**: inner — 4-0 absorbable to vessels, running closure of collecting system (no permanent sutures = stone nidus); outer — capsular sutures over bolster, tension along suture (upward pull tears). Unclamp, check, close Gerota's, drain.

7 • COMPLICATIONS & MANAGEMENT

Injury	Management
SMA	Potentially lethal → vascular repair/graft
Inferior mesenteric artery/vein (IMA/IMV)	Ligation safe (marginal artery of Drummond)
Superior mesenteric vein (SMV)	Never ligate; 6-0 Prolene/vein graft; leave abdomen open (compartment syndrome)
Liver/spleen	Argon beam, spray cautery, glue, pledgeted suture; splenorrhaphy/splenectomy
Duodenum	Haematoma: observe or drain (spare mucosa); tear: 2 layers + omentum + drain + nasogastric tube; don't resect "nonviable-looking" bowel
Pancreas	Glue + drain (amylase); ductal → general surgeon; late: pain, ileus, raised amylase, left renal-bed collection → percutaneous drain, nil by mouth, antibiotics if infected, octreotide
Pleura	Purse-string over 14-Fr catheter, evacuate air on positive-pressure breath; large → chest tube. Late dyspnoea/chest pain → X-ray, oxygen, chest tube

- **Thrombectomy**: air embolism (release caudal clamp first); PE (minimal handling; pulmonary arteriotomy if collapse); bleeding despite control = untied lumbar; **no cell saver** (spreads tumour); liver enzymes peak 2–3 days; post-CPB cardiac/neuro events.
- **Urine leak** (collecting-system entry; large, endophytic, hilar tumours; most appear within 72 h): close big defects primarily; remove drain by 48 h if low. Output >50 mL/24 h → drain creatinine > serum = urine; CT urography only if symptomatic. Drain → radiological drain → if urinoma persists >48 h or symptoms >24 h, retrograde double-J stent + Foley (low-pressure pyelogram). Most resolve 4–6 wk.
- **Bleeding/pseudoaneurysm**: stable → bed rest, serial haemoglobin; significant → **angiography** (CT angiography misses pseudoaneurysm, can't treat) + **selective embolisation**; refractory → re-explore.
- **AVF**: bruit, haematuria, heart failure → observe/coil/stent/surgery.
- **Renal insufficiency**: usually mild, transient; risk with solitary kidney, large resection, long ischaemia; also exclude obstruction, drug toxicity, vascular thrombosis. Supportive care; some need dialysis.

8 • MCQ ONE-LINERS

1. CKD-EPI 2021 is race-free, preferred over MDRD.
2. Thrombus imaging must be ≤1 month old.
3. Postinfarction syndrome lasts up to 72 h.
4. Subcostal flank: poor hilar access, not for PN.
5. Turner-Warwick popularised supracostal incision.
MCQ
6. Right renal artery lies behind renal vein and IVC.
7. SMA arises 1–2 cm above LRV.

8. Left gonadal and adrenal veins drain into LRV.
9. Robson (1969) defined radical nephrectomy.
0. Adrenal metastasis <5%; imaging misses 20-25%.
1. Node short axis ≥ 1.5 cm $\approx$ 50% positive.
2. EORTC LND trial: no OS benefit.
3. Venous thrombus in 5-10% RCC.
4. Level I: ≤ 2 cm above ostium.
5. Level III defined by major, not accessory, hepatic veins.
6. Budd-Chiari contraindicates thrombectomy.
7. Pringle ≤ 60 min, ideally ≤ 20 min.
8. Abort clamp trial: CO $\downarrow > 50\%$ or MAP $\downarrow > 30\%$.
9. DHCA core 16-18 °C.
20. Patch width = diameter $\times$ fraction resected $\times 3$.
21. IVC graft: PTFE 16-20 mm; lifelong aspirin.
22. Bland group D $\rightarrow$ infrarenal IVC resection.
23. Multifocality: papillary $\approx 10\%$, clear cell $\approx 2\%$.
24. Hereditary RCC: operate at 3 cm, except HLRCC.
25. Hyperfiltration if nephron mass $\downarrow > 80\%$.
26. Resection margin 5-10 mm.
27. Hypothermia, mannitol, off-clamp: no RCT benefit.
28. Urine leak: drain creatinine > serum.
29. Pseudoaneurysm: angiography over CT angiography.
30. SMV: never ligate.
31. No cell saver in thrombectomy.

Reading time ≈ 32 min (slow pace)

1 · PRINCIPLES & PATIENT SELECTION

- First laparoscopic nephrectomy: **Clayman, 1991 [MCQ]**. Minimally invasive surgery (MIS) vs open: less pain and blood loss, shorter convalescence, better cosmesis, **equivalent long-term outcomes**. Indications same as open.
- General anaesthesia mandatory**. CO₂ pneumoperitoneum impairs ventilation, venous return and cardiac output. High risk: chronic obstructive pulmonary disease (COPD), intracranial pathology, cardiac disease.
- Severe COPD with hypercarbia: lower insufflation pressure → helium → convert to open.
- Robot: **docking/undocking delays emergency access**; console surgeon may miss the patient slipping → close team communication.
- Obesity: body mass index (BMI) is a poor predictor of difficulty because it does not separate fat from muscle or visceral from subcutaneous fat; **visceral adipose tissue** is the better index. Elderly: benefits apply irrespective of age.
- Prior surgery alters trocar sites and access; it is **not a contraindication**.

KEY POINT The patient must tolerate the **pneumoperitoneum**, not the renal operation — so cardiopulmonary reserve, not age or BMI, decides fitness.

2 · ACCESS & PORT PLACEMENT

Options (laparoscopic or robotic): transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted laparoscopy (HAL), laparoendoscopic single-site surgery (LESS), natural orifice transluminal endoscopic surgery (NOTES).

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar; lens smudging
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	After colon reflection	Direct, esp. posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- TP positioning:** modified flank 30–45°; pad, tape, test tilt; prep whole abdomen for conversion.
- Laparoscopic TP ports:** Veress or Hasson (open) → 3–5 trocars: 10 mm umbilical camera; 5/10 mm midline ~2 cm below xiphoid; 12 mm anterior axillary line (AAL) at umbilicus for staplers. Short patient → 12 mm port to lower midline; obese → shift all lateral.
- Robotic TP:** 3D view, wristed instruments, tremor elimination, motion scaling. Arm spacing **8 cm (Si) / 6 cm (Xi)**; 12 mm low-midline assistant port (5 mm if no stapler needed, e.g. pyeloplasty).
- Whatever the template, ports form a **broad-based triangle**: camera at apex, working ports at base, target at the far point.
- RP:** full flank, table flexed, axillary roll; incision in posterior axillary line midway between 12th rib tip and iliac crest → finger or balloon dissection (glove balloon, 600–800 mL).

KEY POINT RP pitfalls: balloon inflated within muscle → flank hernia; too anterior → peritoneal/colon injury; too posterior → quadratus/psoas bleeding. Bail-out: open the peritoneum and continue TP.

- HAL:** gives tactile feedback and manual retraction; useful for novices, donor nephrectomy and scarred kidneys (xanthogranulomatous pyelonephritis, XGP). Drawbacks: forearm paraesthesia, more wound infection/incisional hernia; hand-port metastases → bag the specimen.

- **HAL emergency role:** hand through an extended port site compresses bleeding while conversion is prepared.
- **LESS / NOTES / single-port (SP) robot:** fewer incisions buy cosmesis and less pain at the cost of triangulation. Operative and oncological outcomes unchanged.

3 · NEPHRECTOMY — OPERATIVE STEPS

ALGORITHM

Reflect colon → find ureter/gonadal vessels on psoas → lift kidney, expose hilum → artery then vein → upper pole (adrenal spared or taken) → ureter and lateral attachments → bag → extract

- **Left colon:** incise white line of Toldt from below the lower pole to above the upper pole; divide lienocolic ligament. Keep the **lateral renal attachments** intact for now — otherwise the kidney rolls over and hides the hilum. The avascular plane lies between bright-yellow mesenteric fat and pale-yellow Gerota fat. Close mesocolic defects (internal hernia).
- **Right:** liver retractor; cold sharp dissection near the duodenum (avoids thermal injury); **Kocher manoeuvre** exposes the inferior vena cava (IVC).
- **Ureter:** follow psoas medially; gonadal vessels are met first, ureter lies just posterior (peristalsis confirms). Do not divide it early — it is the handle for lifting the kidney. Stay above psoas fascia (thigh numbness).
- **Hilum:** lift kidney anterolaterally to put hilum on stretch; artery usually posterior to vein — map on preoperative imaging. **Artery first, then vein** with an endoscopic gastrointestinal anastomosis (GIA) stapler covering the whole vessel. Left renal vein is stapled lateral to the adrenal vein. **Never leave clips in the staple line** — misfire.
- Clips and staplers equally safe. **En bloc hilar stapling** is safe (no arteriovenous fistula found) and saves ~45 min.
- **Upper pole:** adrenal-sparing — open Gerota's above the hilum and peel fat off the upper pole. Extreme fibrosis → subcapsular nephrectomy.
- **Extraction:** intact in a bag via extended port or Pfannenstiel, or morcellation in a **LapSac** (impermeable even after morcellation) under vision, with change of gloves/instruments. Morcellation loses size and capsular/renal-vein staging.
- **Post-op:** diet and activity as tolerated; catheter out once ambulant; no heavy lifting 4–6 weeks if extraction incision.

4 · SIMPLE VS RADICAL NEPHRECTOMY

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple nephrectomy indications:** chronic pyelonephritis (incl. XGP), obstructive/reflux nephropathy, tuberculosis, multicystic dysplastic kidney, renovascular hypertension, symptomatic autosomal dominant polycystic kidney disease (ADPKD).
- Inflammation obliterates planes, so simple nephrectomy can be harder than radical. Predictors: prior pyonephrosis, nephrostomy tube; stone disease has the most complications. Rescue: hand assistance or subcapsular nephrectomy.
- **Radical nephrectomy (RN):** same steps with Gerota's intact. En bloc adrenalectomy → ligate renal vein medial to adrenal vein. RP left: a lumbar vein usually needs division.
- Results: MIS equals open oncologically (5/10 yr) with less morbidity. RP vs TP randomised trials: the only consistent RP advantage is **shorter operative time**.

- **Large tumours (>10 cm):** feasible; use imaging, intraoperative ultrasound, extra trocars or hand port.
- **Lymph node dissection (LND):** staging value only; no overall survival benefit. American Urological Association: remove clinically positive nodes. Template: right paracaval/retrocaval/interaortocaval, left para-aortic; crus to bifurcation.
- **Venous/IVC thrombus** (experts only): early arterial ligation reduces collateral flow and may retract the thrombus → milk thrombus back → staple beyond, or Satinsky clamp on an IVC cuff. Long thrombus: sequential clamping infrarenal IVC → contralateral renal vein → suprarenal IVC; level III needs liver mobilisation. Preoperative angioembolisation helps robotic (not open) surgery.
→ Ch 141

KEY POINT Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery unchanged.

5 · PARTIAL NEPHRECTOMY

- Partial nephrectomy (PN) is the default where feasible: preserved function, equivalent oncology.
- **RENAL nephrometry score** — Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location vs polar lines; predicts complications, warm ischaemia time (WIT), conversion to RN.
- Clinical T1b/T2: PN gives better estimated glomerular filtration rate (eGFR) and less chronic kidney disease (CKD) with equal oncology. **Caveat: tumours >5 cm upstaged to pT3a recur more after PN.** For large tumours the argument for PN is renal function, not oncology.
- **Central** tumour: abuts renal sinus fat/collecting system. **Hilar:** touches renal artery/vein → consider enucleation; use superficial renorrhaphy bites over a bolster to avoid major vessels.
- Multifocal tumours: nephron-sparing (contralateral recurrence risk).
- **Approach choice** depends mainly on tumour location (anterior → TP, posterior → RP) and experience.

ALGORITHM

Intraoperative ultrasound (size, depth) → open Gerota's away from lesion → score capsule → clamp hilum (Satinsky en bloc or bulldogs) → cold-scissor excision → close vessels/collecting system → renorrhaphy → unclamp → drain

- **Trifecta:** negative margin + no major complication within 30 days + **WIT ≤25 min.**
- **Enucleation** follows the pseudocapsule: less blood and function loss, more positive margins, same survival. Use in familial RCC, multifocal disease, solitary kidney, severe CKD.
- **Renorrhaphy:** suturing is most reliable; sliding-clip technique (Lapra-Ty) keeps tension without knots; barbed sutures shorten WIT. **Single-layer** closure gives better function and fewer pseudoaneurysms. Sealants: no randomised benefit.
- **Collecting system** entered in ~60% (large, endophytic tumours). Repair with 2-0/3-0 polyglactin; **always leave a drain** (urinoma). A routine ureteric catheter does not prevent fistula.
- **Ischaemia:** historical warm limit **30 min.** **Preserved parenchymal volume predicts function better than WIT.** Consider cold ischaemia (ice slush, cold perfusion) for solitary kidney, CKD or long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery + vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss

Hilar strategy	Best for	Verdict
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

KEY POINT Why no clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped. Precise excision saves nephrons; minutes of ischaemia matter less.

- Robotic vs laparoscopic PN: fewer conversions, shorter WIT. Robotic vs open: less blood loss, fewer complications.
- **Positive margins (0-7%)** raise recurrence only in high-risk tumours (pT2-3, grade 3-4) → **surveillance, not reoperation.**
- **Port-site recurrence ~0.1%** — due to spillage and technique, not pneumoperitoneum or gas; prevent with impermeable bag.
- Local recurrence after RN (~2%): resection + systemic therapy. Salvage after PN or ablation is hard because planes are lost (no pseudocapsule, fibrosis) → often RN.

6 • OTHER RENAL PROCEDURES

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first — proves the cyst causes pain. Dissect wall, aspirate, excise, biopsy, fulgurate base; collecting-system entry → close + drain. ADPKD: decortication, bilateral nephrectomy, renal denervation
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex; more glomeruli
Nephropexy	Descent >5 cm / 2 vertebrae supine → erect, with pain, reduced Doppler flow or obstruction	Usually right, slim young women; 20% asymptomatic. 2-0 polypropylene capsule to psoas/quadratus fascia; avoid iliohypogastric and genitofemoral nerves
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Laparoscopy sutures the neck closed and obliterates the cavity with fat; endoscopy instead widens the neck
Nephrolysis	Chyluria, loin pain-haematuria	Lateralise by cystoscopy; hilar lymphatic ligation ± omental wrap
Ablation	≤3 cm, comorbid, solitary, hereditary RCC	Percutaneous now preferred. Cryo: two freeze-thaw cycles, iceball 0.5-1 cm beyond tumour on real-time ultrasound. Radiofrequency ablation (RFA): 60-100°C (<105°C), 1 cm margin, impedance-guided; more retreatment

7 • COMPLICATIONS & MANAGEMENT

- Overall **~15%**, mostly postoperative; PN > RN > simple. Learning curve ~20 laparoscopic nephrectomies. Robotic PN halves open conversion vs laparoscopic.
- Prior surgery: place Veress/first trocar away from scars, use Hasson access, or go retroperitoneal.
- **Vascular injury is the commonest complication**; major injury <1/1000 but fatal in 8-17%. Occurs at **access** (Veress, first trocar) or hilar dissection. Device failure: staplers > non-locking titanium clips > locking clips; stapler errors are mostly operator (stapling over clips, tissue too thick, early stop).
- **Left RN pitfall**: superior mesenteric artery (SMA) mistaken for renal artery → bowel necrosis; needs open reconstruction.

- Venous ooze: gauze pressure for minutes. Brisk arterial bleed: hand through a lower midline incision to compress. **Never clip or suture blindly.**

ALGORITHM

Major bleed: locate → temporary control (grasper/pressure) → inform anaesthetist (blood, open set) → clear field, raise pneumoperitoneum or add trocar → definitive clip/suture → else convert and call vascular surgeon

- End of case: **lower insufflation pressure** to unmask venous bleeding.
- Post-op haemorrhage (hypotension, tachycardia, falling haematocrit) → computed tomography (CT); arterial source → angioembolisation. **Haematuria ~day 12 after PN** → pseudoaneurysm/arteriovenous fistula → selective embolisation (endophytic, posterior tumours at risk).
- **Urine leak** after PN (0.8–10%): percutaneous drainage → if persistent or distal obstruction, ureteric stent + catheter, or nephrostomy.
- **Unrecognised bowel injury** (<1%, mortality ~3%): thermal energy causes about half and presents later; 32% occur at access. Classic picture: **pain at a single trocar site**, distension, diarrhoea, low-grade fever, **normal or low white count**, bowel sounds present. First test: CT with oral contrast. Stable → laparoscopic repair; unstable/gross spillage → laparotomy. Recognised superficial thermal injury → imbricate; devitalised segment → resect.
- **Solid organs:** spleen (left traction) → haemostatic agents/argon → splenorrhaphy → splenectomy; pancreas → suture ± drain or distal pancreatectomy.
- **Diaphragm** (upper pole): sudden rise in airway pressure → purse-string, suction tube withdrawn on full inspiration, chest X-ray; drain large pneumothorax.
- † CO₂ gas embolism: sudden hypotension, desaturation, fall in end-tidal CO₂ → stop insufflation, left-side-down head-down, aspirate via central line.
- **Neuromuscular** (2.7%): rhabdomyolysis with long operations in full flank — buttock/thigh pain, raised creatine kinase, myoglobinuria → hydration, correct acidosis. Axillary roll prevents brachial plexus injury.
- **Thromboembolism:** lower than open; low risk → compression devices, high risk → heparin. Also port-site hernia, ileus.

KEY POINT MIS complications are the open ones in altered form and timing: the lethal ones — access vessel injury, thermal bowel injury — happen out of view and present late or atypically.

8 • MCQ ONE-LINERS

- | | |
|--|--|
| 1. First laparoscopic nephrectomy: Clayman 1991; first laparoscopic PN: Winfield 1993. | 10. >5 cm tumour upstaged to pT3a: more recurrence after PN than RN. |
| 2. Better predictor of difficulty than BMI: visceral adipose tissue. | 11. Cryo iceball 0.5–1 cm beyond tumour; two freeze-thaw cycles. |
| 3. Robot arm spacing: 8 cm Si, 6 cm Xi. | 12. RFA 60–100°C; >105°C causes vaporisation; no real-time imaging. |
| 4. RP vs TP radical nephrectomy: consistent RP advantage is shorter operative time. | 13. Device failure: staplers > non-locking clips > locking clips. |
| 5. En bloc hilar stapling: no arteriovenous fistula; saves ~45 min. | 14. Bowel injury mortality ~3%; thermal causes about half. |
| 6. Trifecta WIT ≤25 min; historical warm ischaemia limit 30 min. | 15. Unrecognised pneumothorax after laparoscopic nephrectomy ~0.3%. |
| 7. Collecting system entered in ~60% of laparoscopic PN. | 16. Neuromuscular injury in 2.7% of laparoscopic urological surgery. |
| 8. Positive margins 0–7% of MIS PN; matter only in high-risk tumours. | 17. Learning curve ~20 laparoscopic nephrectomies. |
| 9. Port-site recurrence ~0.1% of laparoscopic cancer operations. | 18. Commonest MIS complication: vascular injury. |
| | 19. Robotic PN morbidity plateaus at 18–20 hospital cases/year. |

!0. Nephroptosis: >5 cm or 2 vertebral bodies;
asymptomatic in 20% of women.

Reading time ≈ 21 min (slow pace)

1 · RATIONALE & INDICATIONS

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do. So treatment minimises competing-risk harm rather than maximising radicality. Small renal mass (SRM) biology/biopsy → *Ch 137*
- **Partial nephrectomy (PN) = gold standard** for clinical T1a (cT1a). Thermal ablation (TA) – cryoablation (CA), radiofrequency (RFA), microwave (MWA) – is the alternative.
- **Indications:** high surgical risk; solitary kidney; bilateral tumours; hereditary syndromes, e.g. von Hippel-Lindau (VHL), where repeat treatments are expected.
- No dissection/renorrhaphy → faster recovery, fewer complications; renal function ≈ or better than PN.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a ≤3 cm , esp. comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	TA not routinely ≥3 cm ; CA not ≥4 cm ; for frail/comorbid
National Comprehensive Cancer Network (NCCN) 2022	Only one allowing cT1b (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · SELECTION & WORK-UP

ALGORITHM

Counsel (age, comorbidity, CKD risk, size/location) → biopsy → re-counsel with histology → PN / surveillance / percutaneous ablation

- **AUA: biopsy all**, preferably at a **separate earlier session**: non-diagnostic biopsy can be repeated, fewer benign masses ablated, better-informed consent. Post-ablation biopsy only for a new enhancing lesion.
- Urine culture; treat urinary tract infection (UTI) first – colonised urine (ileal conduit) risks abscess.
- Complication predictors: RENAL nephrometry score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (~3×), (MC)² (CA only).
- Day-case: ≤3 cm, low complexity, no anticoagulation, few comorbidities; else 23-h stay.

3 · MECHANISMS

- Heat and cold share two zones: **central** direct necrosis, and **peripheral sublethal** apoptosis, ischaemia and immune response.
- **Cryo kill:** freezing – extracellular ice dehydrates cells, then intracellular ice ruptures them; thawing – water floods back → lysis. Plus vascular thrombosis and immune activation (released antigens → dendritic cells → T cells).
- Kidney dies at –19.4 °C, tumour may need –50 °C → target **≤–40 °C**. Iceball edge is 0 °C; <–20 °C only 3 mm inside it → iceball **5–10 mm beyond tumour**.
- Heat: 40–45 °C sublethal; **≥60 °C** coagulative necrosis; ≥100 °C vaporisation/charring. Lethal: **70 °C for 1 min**.
- **Heat sink:** adjacent large vessels carry heat/cold away → surviving perivascular tumour.

4 · CHOICE OF ENERGY

	Cryoablation	RFA	MWA
Physics	Argon expansion cools (Joule-Thomson); helium warms	450–1200 kHz current → ionic friction	915 MHz/2.45 GHz → dielectric hysteresis
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	Double freeze-thaw (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	Yes	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	Least
Zone visible	Iceball	No – impedance/temp	Gas bubbles on US
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

KEY POINT Why RFA is self-limiting: charring raises impedance and halts current, so temperature is capped at 105 °C and electrodes are cooled. MWA *radiates* energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- Double freeze-thaw: thaw itself kills and a second freeze enlarges necrosis. 10 min is a compromise – 5 min under-treats, 15 min fractures tissue.
- Thaw: passive, or active (helium). Common hybrid – passive between cycles, active at the end so bleeding is addressed sooner.
- RFA: temperature- vs impedance-controlled generators equivalent; multitime electrodes best. Wet RFA (saline infusion) → larger but less controlled zone; bipolar → elliptical zone, not adopted.
- MWA antenna cooling protects the track and allows higher power; RFA cooling prevents charring.

5 · TECHNIQUE

- **Percutaneous = standard.** Laparoscopic is equally effective but more morbid and costly – keep for prior scarring or failed hydrodisplacement.
- **General anaesthesia preferred:** controlled respiration improves probe placement; high-frequency jet ventilation (~120/min, small tidal volumes) reduces renal motion. Conscious sedation only if intubation not tolerated.

ALGORITHM

IV antibiotics → prone/flank on computed tomography (CT) gantry → ultrasound (US), CT or magnetic resonance imaging (MRI)-guided placement → non-contrast CT confirms position → ablate → contrast CT (completeness, haematoma)

- **Hydrodisplacement:** needle into plane between tumour and bowel/ureter/organ → saline, or 5% dextrose (**D5W for RFA**, †saline conducts current) ± contrast; aim e.g. colon >1 cm. Caution in cardiac failure.
- **Pyeloperfusion** (central tumours): ureteral catheter irrigated at the **opposite temperature** – cold for RFA/MWA, warm for CA.
- **CA:** multiple probes **1–2 cm apart**; remove only after full thaw, without resistance – early removal fractures tumour and bleeds.
- **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; ablate track only past Gerota fascia (spare nerves).

- **MWA:** antennas tangential, never pointing into collecting system; 1 antenna ≤ 2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm. Cooled antenna tip anchors it. Poor renal function → check with contrast-enhanced US instead of contrast CT.

6 • FOLLOW-UP, SUCCESS & SALVAGE

- **Success = complete loss of contrast enhancement** on CT/MRI. The tumour stays in situ, so an imaging margin replaces the pathological margin – hence biopsy first and long surveillance.
- **Incomplete ablation:** enhancement on the *first* scan → repeat ablation.
- **Local recurrence:** initially non-enhancing, *later* enlarges/enhances → biopsy; surveillance, re-ablation or excision.
- **Enlargement after any modality = recurrence until proved otherwise.**
- Appearance: CA shrinks >50% in year 1; RFA minimal shrinkage, **fibrotic halo**; MWA low-attenuation zone that keeps contracting; MRI T1-hypointense, non-enhancing.
- **AUA:** CT/MRI with and without contrast + chest imaging at **6, 12, 24, 36, 48, 60 months**; later shared decision. US not routine.
- Salvage: re-ablation, PN, radical nephrectomy (RN), or stereotactic radiotherapy if comorbid.

7 • ONCOLOGICAL OUTCOMES VS PN

- No randomised comparison. Local recurrence-free survival (LRFS) worse after TA but **equal once re-ablation counted; metastasis-free and cancer-specific survival (MFS, CSS) equal**; overall survival lower (older, comorbid).
- **Size predicts failure:** RFA worse ≥ 3 cm; T1b worse than T1a. No energy superior; guidelines accept all three – choose by institutional preference and tumour.
- CSS: CA 80–100% at 10 years, RFA $\approx 98\%$; overall survival 75–85% at 5 years (comorbidity, not cancer).

KEY POINT Why comparisons are biased: benign tumours and under-graded biopsies flatter TA; occult pT3a and growth-selected tumours penalise it. European Organisation for Research and Treatment of Cancer (EORTC) 30904 (PN vs RN, ≤ 5 cm) had only 12 RCC deaths at 9.3 years – trials need huge cohorts.

8 • COMPLICATIONS & MANAGEMENT

- Major (Clavien ≥ 3) **$\leq 5\%$** , fewer than PN; percutaneous < laparoscopic. Pain minimal.

Complication	Prevention	Management
Haemorrhage - commonest ; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Bleeding at placement → start ablating (RFA/MWA coagulate); persistent → selective embolisation; wall haematoma → serial imaging/haematocrit
Urinoma, fistula, pelviureteric junction (PUJ)/ureteric stricture	Stent + pyeloperfusion; CA for central; tangential antennas	Stent; drain large urinoma
Bowel/organ (tumour <1 cm away)	Hydrodisplacement, positioning	Surgical consult; colonephric fistula → stent + low-residue diet; persistent → diversion; peritonitis → explore
Pneumothorax (above 12th rib)	—	Aspirate at end; larger → blood patch; chest tube rarely
Nerve injury / RFA pad burn	Tumour falls away from wall; pads level on posterior thighs	Self-limiting
Infection <1%, may be lethal; abscess/drainage at 1–6 months	Urine culture, prophylaxis	†Drain + antibiotics

9 • NEWER MODALITIES (INVESTIGATIONAL)

- **High-intensity focused ultrasound (HIFU):** entirely extracorporeal (no seeding, bleeding or leak), thermal + cavitation; ablate-and-resect studies showed incomplete ablation (breathing, acoustic interference).
 - **Histotripsy:** focused ultrasound, mechanical (non-thermal); no clinical data yet.
 - **Stereotactic radiotherapy (SBRT/SABR):** no anaesthesia, anticoagulation continued; but no imaging endpoint (little change after treatment), viable attenuated tumour on biopsy, lower-than-expected CSS. Possible role: salvage after TA/PN in comorbid patients.
 - **Irreversible electroporation (IRE):** non-thermal pores; spares matrix and vessels; poor oncological control.
-

10 • MCQ ONE-LINERS

- | | |
|---|--|
| 1. AUA: TA for cT1a $\leq$ 3 cm; only NCCN allows cT1b. | 13. Lowest complication rate: MWA (network meta-analysis). |
| 2. Argon cools, helium warms (Joule-Thomson). | 14. CSS and MFS equal TA vs PN. |
| 3. Cryo target ≤ -40 °C; iceball 5-10 mm beyond tumour. | 15. Histotripsy is non-thermal; HIFU thermal + cavitation. |
| 4. MWA works by dielectric hysteresis; RFA by ionic friction. | 16. IRE preserves extracellular matrix and vessels. |
| 5. >70 °C for 1 min lethal to RCC. | 17. Normal renal parenchyma dies at -19.4 °C. |
| 6. Only RFA needs grounding pads and is limited by charring. | 18. <-20 °C reached only 3 mm inside iceball edge. |
| 7. MWA least susceptible to heat sink. | 19. RFA 450-1200 kHz; MWA 915 MHz or 2.45 GHz. |
| 8. Only CA shows a visible real-time zone. | 20. D5W, not saline, for hydrodisplacement with RFA. |
| 9. After RFA: fibrotic halo; after CA: $>50\%$ shrinkage in year 1. | 21. Success = complete loss of enhancement on CT/MRI. |
| 10. Pyeloperfusion temperature opposes the ablation. | 22. AUA follow-up imaging: 6, 12, 24, 36, 48, 60 months. |
| 1. CA preferred for central tumours. | 23. Biopsy before ablation, ideally at a separate session. |
| 2. Commonest complication: haemorrhage; major $\leq 5\%$. | 24. Laparoscopic ablation: prior scarring or failed hydrodisplacement. |

144 Treatment of Advanced Renal Cell Carcinoma

Reading time ≈ 36 min (slow pace)

1 • OVERVIEW

- Renal cell carcinoma (RCC) is a family of distinct cancers. About **1/3** present with synchronous metastases; a further **20-40%** of localised tumours later metastasise. Metastatic RCC (mRCC) 5-yr survival ≈ **12%**; 10-yr <5%.
- Three treatment eras: cytokines → single-agent targeted therapy → **immune checkpoint inhibitor (immuno-oncology, IO) combinations**.
- Clear cell RCC (ccRCC) is the main target of all these drugs; non-clear cell RCC (nccRCC) has little prospective evidence.
- Pathology, staging and nomograms for localised disease, and adjuvant therapy (KEYNOTE-564, S-TRAC) → *Ch 137*.

2 • PROGNOSTIC MODELS

- Good prognosis: long interval from diagnosis to metastasis, few metastatic sites. Bad: poor performance status (PS), lymph-node or liver metastases.
- MSKCC (Memorial Sloan-Kettering Cancer Center, Motzer) model:** cytokine era, 670 patients mostly on interferon-α (IFN-α). **IMDC (International Metastatic RCC Database Consortium, Heng) model:** re-derived in patients on first-line vascular endothelial growth factor (VEGF)-targeted therapy; validated in 849.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	>10 mg/dL	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	>1.5 × ULN	—
Time factor	No prior nephrectomy	<1 yr diagnosis → systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1-2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

KEY POINT Both models score host reserve (PS, anaemia), tumour-driven systemic effects (calcium, LDH, neutrophils, platelets) and pace of disease. Three factors are shared; IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and a short interval. Poor risk = ≥3 factors in both. The IMDC risk group now **decides the first-line regimen**.

3 • BIOLOGY AND DRUG TARGETS

- VHL (von Hippel-Lindau) gene**, chromosome 3p: classic tumour suppressor (second hit needed). Its protein, with elongins B/C and cullin-2 (CUL2), tags **hypoxia-inducible factor (HIF)-α** for ubiquitin-mediated degradation after prolyl hydroxylation in normoxia.
- When VHL is mutated or silenced (promoter hypermethylation) – as in most sporadic ccRCC – HIF-α accumulates even in normoxia and constitutively drives VEGF, platelet-derived growth factor (PDGF), transforming growth factor-α (TGF-α), glucose transporter-1 (GLUT-1) and erythropoietin. This makes the VHL-HIF-VEGF axis the logical drug target.

- Each targeted drug blocks one consequence of VHL loss: the ligand (bevacizumab), its receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mammalian target of rapamycin [mTOR] inhibitors) or HIF-2 α itself (belzutifan). Resistance through MET, AXL and fibroblast growth factor receptor (FGFR) explains why later TKIs are broader.
- **Checkpoints:** cytotoxic T-lymphocyte antigen-4 (CTLA-4; a CD28 homologue binding B7) and programmed death-1 (PD-1, on T cells) with its ligands PD-L1/PD-L2 (tumour, macrophages) switch off antitumour T cells. Checkpoint inhibitors release this brake rather than stimulating immunity nonspecifically, so toxicity is autoimmune and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory (sunitinib reduces regulatory T cells; sunitinib/sorafenib suppress tumour-associated macrophages) – the rationale for IO + TKI combinations.
- ccRCC is chemoresistant (response 5.5–6%; multidrug resistance [MDR] inhibitors failed) and hormone-insensitive (medroxyprogesterone ~2%).

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A ligand; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR- β , raf-1	Rash, hand-foot syndrome (HFS); now rarely used
Sunitinib	VEGFR2, PDGFR- β , c-KIT, FLT3; 50 mg, 4 wk on / 2 off	Diarrhoea, HFS, fatigue, HTN, thrombocytopenia, hypothyroidism
Pazopanib	VEGFR, PDGFR, KIT	Hepatotoxicity ($\uparrow$ alanine aminotransferase (ALT)) ; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR- β	Third line
Cabozantinib	VEGFR, MET, AXL	Fatigue, diarrhoea, palmar-plantar erythrodysesthesia; frequent dose reduction
Lenvatinib	VEGFR1–3, FGFR1–4 , PDGFR- α , RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; IV weekly	Mucositis, rash, $\uparrow$ glucose, $\downarrow$ phosphate, $\uparrow$ cholesterol, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	HIF-2α inhibitor ; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis/enteritis, hypophysitis, vitiligo
Avelumab	Anti-PD-L1 ; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · SURGERY IN METASTATIC DISEASE

Cytoreductive nephrectomy (CN)

- **Rationale:** rare spontaneous regression of metastases after nephrectomy (<1%, often transient); bulky primaries suppress T-cell function; cytokines rarely shrink the primary. Nephrectomy alone does not alter outcome – it only helps as a platform for systemic therapy.
- **Drawback:** perioperative morbidity and rapid progression mean a significant proportion never reach systemic therapy (only ~half in the high-dose interleukin-2 [IL-2] series; <1/4 if ECOG [Eastern Cooperative Oncology Group] PS 2, multiple sites, brain or liver metastases).
- **Cytokine era:** SWOG 8949 and EORTC trials – CN + IFN- α improved OS without changing response rate, so benefit was not via better cytokine response.
- **TKI era:** CARMENA – sunitinib alone noninferior to CN + sunitinib in MSKCC intermediate/poor risk. SURTIME – deferring CN until after sunitinib was safe, OS favoured deferral.

- **CARMENA caveats:** ~45% poor risk (not normally offered CN); ~25% of CN arm off-protocol; 17% of sunitinib arm had later CN; per-protocol noninferiority lost; investigators could exclude low-burden patients; stopped early, underpowered; sunitinib no longer standard. It does **not** show harm in fit, low-burden, favourable patients.
- **Unlikely to benefit:** poor PS, comorbidity, rapidly progressive disease, brain metastases, poor risk. Role in nccRCC unclear. Authors offer CN to **well-selected** ccRCC patients.

ALGORITHM

mRCC → histology + IMDC risk + PS + metastatic burden

Fit, low burden, favourable features, resectable primary → consider upfront CN → systemic therapy

Intermediate/poor risk or high burden → systemic therapy first → deferred CN in responders

KEY POINT CN helps only patients fit enough, and with disease slow enough, to reach effective systemic therapy. When systemic therapy is active, operating first on poor-risk patients mainly delays it – hence selection and deferral.

Metastasectomy

- No randomised trial; retrospective series. Isolated **pulmonary** metastases are most often resected with curative intent; 5-yr survival **35-50%** after complete resection of limited disease.
- Favourable: **complete resection**, solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm, metachronous. Complete resection still helps with multiple lesions.
- Every predictor is also a marker of indolent disease, so the operation selects good biology as much as it treats it; still a reasonable, widely used practice.

Palliative surgery and radiation

- **Palliative nephrectomy:** intractable pain, haematuria, constitutional symptoms, paraneoplastic syndromes (hypercalcaemia, erythrocytosis, thrombocytosis, HTN). Try non-surgical options first: medical control of pain and hypercalcaemia; **angioembolisation for haematuria**. Only ~half of hypercalcaemic patients improve after resection, so purely palliative nephrectomy is infrequent.
- **Palliative metastasectomy:** solitary brain metastasis; lesions in weight-bearing bones or joints; vertebral metastases with impending cord or root compromise – usually combined with **radiation** and/or systemic therapy.

5 · SYSTEMIC THERAPY FOR CCRCC

Cytokines

	IFN- α	High-dose IL-2
Response	10-15%	15-20%
Durable complete response (CR)	<2%	7-9% ; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	600,000-720,000 IU/kg IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	Vascular leak: hypotension, third-spacing, respiratory and multiorgan failure; early mortality 2-5%
Current role	Not used first-line	Carefully selected fit patients

- High-dose IL-2 survives because it alone produces durable complete remissions. Low-dose regimens gave fewer durable CRs in two randomised trials, so only high dose is recommended. IL-2 toxicity is controlled by careful selection, intensive monitoring, early IV fluids, vasopressors and antibiotics.
- IL-2 combinations (IFN, 5-fluorouracil, tumour-infiltrating lymphocytes) add toxicity, not survival. Best IL-2 candidates: clear cell, good PS, prior nephrectomy; worst: >1 metastatic site, metastasis within 1 yr, liver metastases.

First-line therapy by IMDC risk

IO/TKI or IO/IO combination is preferred by both the National Comprehensive Cancer Network (NCCN) and the European Association of Urology (EAU). Every first-line approval since 2018 is a checkpoint combination tested against sunitinib.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	Active surveillance , axitinib, high-dose IL-2
Intermediate / poor	Same three + ipilimumab + nivolumab	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, temsirolimus

KEY POINT Choosing: IO/TKI gives higher response and PFS with better tolerance; IO/IO (ipilimumab + nivolumab) gives deeper, durable responses (CR 9%) at the cost of autoimmune toxicity – they have never been compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk because favourable-risk patients did better on sunitinib in CheckMate 214. Avelumab + axitinib is approved but not NCCN-preferred (no OS benefit). PD-L1 testing does not select patients for any regimen. Single-agent TKI or temsirolimus is the fallback when IO is contraindicated.

Subsequent therapy - switch mechanism

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	Nivolumab or cabozantinib (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, belzutifan , cabozantinib, lenvatinib + everolimus, tivozanib

- Whichever axis was used first, the next line attacks the other (or an unused VEGF target). IO re-challenge after IO is only "useful in certain circumstances".
- Cabozantinib works after VEGF failure because MET and AXL are HIF-driven escape routes. Belzutifan: after both PD-1/PD-L1 and VEGF TKI. Tivozanib: after ≥ 2 prior therapies. Everolimus alone is supplanted, but used with lenvatinib.
- Failed combinations: sunitinib + mTOR inhibitor and nivolumab + sunitinib/pazopanib (too toxic); bevacizumab + mTOR inhibitor (RECORD-2, INTORACT – no gain).

Toxicity and its management

- **Ipilimumab + nivolumab:** grade 3–4 events **46%**; immune-related events needing high-dose glucocorticoids 35%; permanent discontinuation 22%. The chosen dose (nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly $\times 4$) was the least toxic in CheckMate 016. Nivolumab alone is better tolerated than everolimus (grade 3–4: 19% vs 37%).
- $\uparrow$ Immune-related adverse events: hold IO and start corticosteroids for grade ≥ 2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥ 4 weeks; grade 4 usually means permanent discontinuation.
- $\uparrow$ Endocrinopathies (hypothyroidism, hypophysitis) are managed with hormone replacement; check thyroid function and cortisol if fatigue or hypotension develops.
- **TKIs:** HTN, diarrhoea, HFS, fatigue, hypothyroidism – managed by dose reduction/interruption; few stop permanently. Avoid pazopanib if hepatotoxicity risk.
- **mTOR:** monitor glucose, lipids, phosphate; watch for pneumonitis.

6 · NON-CLEAR CELL RCC

- 15–25% of kidney cancers; no agent of proven efficacy, so ccRCC drugs are used with modest activity. **Offer trial enrolment to all.**

- **Papillary:** sunitinib response 5–10%; erlotinib (EGFR inhibitor) modest. Temsirolimus beat IFN in the non-clear cell subgroup (~20%) of its poor-risk trial. Everolimus (RAPTOR) modest; sunitinib vs everolimus – neither clearly better, trend to sunitinib. Chromophobe: trend to longer PFS on everolimus.
- **Hereditary papillary RCC (HPRC):** germline activating **MET** mutation (chromosome 7) → bilateral multifocal type 1 papillary. MET is the hepatocyte growth factor (HGF) receptor; mutation makes it constitutively active. Sporadic papillary: somatic MET ~13%, **trisomy 7 (HGF + MET) in >2/3**. Foretinib (MET + VEGFR2): about half of germline MET carriers responded vs few others; selective MET inhibitors (savolitinib, crizotinib) active in MET-driven tumours.
- **Hereditary leiomyomatosis and RCC (HLRCC):** germline fumarate hydratase (FH) loss → aggressive type 2 papillary. Fumarate causes VHL-independent HIF activation, and Krebs-cycle failure makes cells glycolysis-dependent and sensitive to glucose deprivation – hence **bevacizumab + erlotinib** to restrict glucose delivery (partial response ~2/3).
- **Collecting duct:** resembles urothelial carcinoma → **gemcitabine + carboplatin. Sarcomatoid:** gemcitabine-based chemotherapy promising (small series); more PD-1-positive infiltrates.

KEY POINT In nccRCC the hereditary syndrome names the driver, and the driver names the drug: VHL → VEGF/mTOR/HIF-2α; MET → MET inhibitors; FH → glucose restriction.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	Basis of CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs 18.4 mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor risk
SURTIME	Immediate vs deferred CN	OS 15.0 vs 32.4 mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Comparator of all later trials
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; differ in toxicity; quality of life and patient preference (PISCES) favoured pazopanib
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS ↑, OS not	Bevacizumab little used
Temsirolimus phase III	Temsirolimus vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	Everolimus vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Now supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer than everolimus	Approval of lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF therapy	PFS HR 0.58; OS HR 0.70	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib

Trial	Design	Result	Take-home
TIVO-3	Tivozanib vs sorafenib, after 2-3 lines	PFS HR 0.73; OS same	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (medians equal – curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, HR 0.73 ; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS ↑; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS ↑; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, HR 0.39 ; response 71%	Largest PFS/response effect

8 • MCQ ONE-LINERS

- MSKCC: LDH >1.5 × ULN, calcium >10 mg/dL, no prior nephrectomy.
- IMDC adds neutrophilia, thrombocytosis; drops LDH.
- Poor risk = ≥3 factors; IMDC poor-risk median OS 7.8 mo.
- VHL: chromosome 3p; complexes with elongin B/C and CUL2.
- HPRC: MET, chromosome 7, type 1 papillary.
- Sporadic papillary RCC: trisomy 7 in >2/3.
- HLRCC: fumarate hydratase; type 2 papillary; bevacizumab + erlotinib.
- Most selective VEGFR TKI: axitinib.
- Only anti-PD-L1 used in RCC combinations: avelumab.
- Lenvatinib uniquely inhibits FGFR1-4; cabozantinib inhibits MET and AXL.
- Sunitinib: 50 mg, 4 weeks on, 2 off; causes hypothyroidism.
- Pazopanib: raised ALT; COMPARZ noninferior to sunitinib.
- High-dose IL-2: 600,000-720,000 IU/kg 8-hourly, max 15 doses.
- IL-2 dose-limiting toxicity: vascular leak syndrome.
- IL-2 durable CR 7-9%; only agent approved on phase II response.
- Temsirolimus: IV weekly; only mTOR inhibitor with OS benefit.
- Temsirolimus is listed for intermediate/poor risk only; active surveillance for favourable only.
- CheckMate 025 primary endpoint OS, HR 0.73.
- CheckMate 214: CR 9% vs 1%.
- CLEAR PFS HR 0.39 – largest among IO combinations.
- JAVELIN Renal 101: only combination without OS benefit.
- First-line regimens limited to intermediate/poor risk: ipilimumab + nivolumab, cabozantinib, temsirolimus.
- PD-L1 status does not select first-line therapy.
- Belzutifan targets HIF-2α; approved after IO and TKI, and for VHL disease.
- Chemotherapy response in ccRCC: 5.5-6%.
- Collecting duct carcinoma: gemcitabine + carboplatin.
- Spontaneous regression after nephrectomy: <1%.
- Metastasectomy 5-yr survival after complete resection: 35-50%.
- SURTIME: deferred CN OS 32.4 vs 15.0 mo.
- CARMENA: ~45% MSKCC poor risk; sunitinib alone noninferior.
- Haematuria from unresectable primary: angioembolisation.
- Autoimmune events with ipilimumab predicted response.

★ Putting it together: the big treatment choices

Reading time ≈ 10 min (slow pace) · Integrates Ch 136–144 · use these tables as the skeleton of long-essay answers

A · SMALL RENAL MASS (SRM): WHICH TREATMENT?

Four options exist for a localised solid renal mass: active surveillance (AS), thermal ablation (TA), partial nephrectomy (PN) and radical nephrectomy (RN). The choice balances cancer control against kidney function and the patient's fitness. Renal mass biopsy (RMB) is used when its result would change this choice.

	Active surveillance	Thermal ablation	Partial nephrectomy	Radical nephrectomy
Best for	Elderly, frail, life expectancy <5 y; solid <2 cm or mainly cystic	cT1a ≤3 cm , comorbid; biopsy first	cT1a standard of care; imperative in solitary kidney, bilateral, hereditary tumours or chronic kidney disease (CKD)	Only when all three: high complexity, no CKD/proteinuria, expected estimated glomerular filtration rate (eGFR) ≥45
Key numbers	Growth 0.1–0.3 cm/y; metastasis <2%. Intervene if >3 cm or growth >5 mm/y	Cryoablation: 2 freeze–thaw cycles; radiofrequency (RFA) / microwave (MWA): heat >60–70 °C	"Trifecta": negative margin, no complications, warm ischaemia ≤25 min. Margin width is immaterial	Predicted new GFR ≈ GFR × contralateral split function × 1.25
Weakness	Needs adherence to imaging	More local recurrence (treated by re-ablation); no final pathology	Urine leak, bleeding, pseudoaneurysm	CKD in 65% after RN vs 20% after PN
See	Ch 137	Ch 143	Ch 137, 141, 142	Ch 137, 141, 142

B · RENAL LESION AT A GLANCE

Lesion	Imaging clue	Genetics / immunohistochemistry	Default action
Simple cyst	≤20 Hounsfield units (HU), no enhancement	—	Nothing
Complex cyst	Bosniak v2019 class	—	Size-based: ≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Macroscopic fat ≤–20 HU; fat-poor AML is the trap	TSC2/TSC1; HMB-45, Melan-A positive	>6 cm, aneurysm >5 mm or bleeding → selective embolisation; tuberous sclerosis → everolimus
Oncocytoma	Central scar, segmental enhancement inversion	KIT+, CK7 single cells; Birt-Hogg-Dubé	Biopsy cannot exclude chromophobe RCC → surveil or PN
Clear cell renal cell carcinoma (ccRCC)	Hypervascular, heterogeneous	3p loss / VHL → HIF-2α; CA-IX	PN, RN, TA or AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple chromosome losses; diffuse CK7	As RCC; good prognosis

Lesion	Imaging clue	Genetics / immunohistochemistry	Default action
Lymphoma / metastasis	Infiltrative, hypovascular, adenopathy	—	Biopsy; do not operate
Upper tract urothelial carcinoma (UTUC)	Filling defect, ~46 HU, positive cytology	Microsatellite instability → Lynch	Risk-stratify → kidney-sparing or nephroureterectomy

C · UTUC: KIDNEY-SPARING MANAGEMENT (KSM) VS RADICAL NEPHROURETERECTOMY (RNU)

	KSM (ureteroscopic, percutaneous, segmental resection)	RNU with bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low-grade cytology and biopsy, no invasion on computed tomography urography (CTU). Imperative: solitary kidney, CKD, bilateral disease	High risk: ANY of multifocal, >2 cm, high grade, hydronephrosis, previous cystectomy, variant histology
Extras	Ureteroscopy at every surveillance visit; mitomycin gel (UGN-101) for low grade; about 3× more re-interventions	Within 12 weeks; template lymph node dissection; single post-operative dose of intravesical chemotherapy
Systemic	—	Neoadjuvant platinum if post-op creatinine clearance will fall <50; adjuvant platinum (POUT) for ≥pT2 or node-positive

D · OPEN VS MINIMALLY INVASIVE (LAPAROSCOPIC/ROBOTIC) KIDNEY SURGERY

Choose open (Ch 141)	Choose laparoscopic / robotic (Ch 142)
Locally advanced RCC (especially inferior vena cava thrombus needing bypass); cytoreductive setting; anatomically complex tumours; significant previous abdominal surgery. Trauma: midline incision with early vascular control	Standard for RN of organ-confined tumours and most PNs: less pain, blood loss and hospital stay, same cancer outcomes. The retroperitoneal route gives shorter operating time

E · METASTATIC RCC IN FOUR STEPS

ALGORITHM 1. Risk group (IMDC, International Metastatic RCC Database Consortium; 6 factors): Karnofsky <80%, haemoglobin < lower limit of normal, calcium, neutrophils or platelets above upper limit of normal, <1 year from diagnosis to systemic therapy → 0 favourable · 1–2 intermediate · ≥3 poor. 2. First line: immunotherapy (IO) + tyrosine kinase inhibitor (TKI) (pembrolizumab + axitinib, nivolumab + cabozantinib, pembrolizumab + lenvatinib) for any risk; ipilimumab + nivolumab for intermediate/poor risk. 3. Cytoreductive nephrectomy: upfront only if fit, low tumour burden and favourable risk; otherwise systemic therapy first and deferred nephrectomy in responders (CARMENA, SURTIME). 4. Later lines: switch mechanism; belzutifan (HIF-2α inhibitor) after both IO and TKI.
--